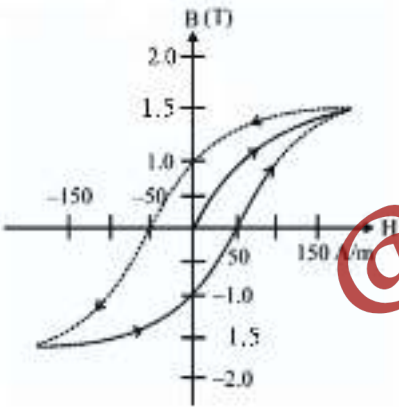
PHYSICS

1) Given below are two statements :
Statement-I: The magnetic force acting on a moving charge can never change speed of the charge.
Statement-II: An electric field can only speed up or slow down a moving charge but can never deflect the charge.
In the light of the above statements. choose the most appropriate answer form the options given below.

(1) Both statement I and statement II are incorrect
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct
(4) Both statement I and statement II are correct

2) The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are :



@examlix

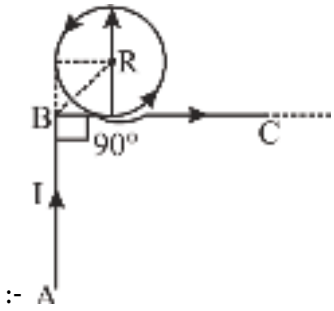
(1) 150 A/m, 1.0 T and 1.5 T
(2) 1.0 T, 50 A/m and 1.5 T
(3) 1.0 T, 50 A/m and 1.0 T
(4) 1.5 T, 50 A/m and 1.0 T

3) A proton carrying 1 MeV kinetic energy is moving in a circular path of radius R in uniform magnetic field. What should be the energy of an α-particle to describe a circle of same radius in the same magnetic field.

(1) 0.5 MeV
(2) 4 MeV
(3) 2 MeV
(4) 1 MeV

4) The magnetic field at the centre of the circular loop as shown in figure, when a single wire is bent to form a circular loop and also extends to form straight sections, is (Assume BC is straight section)

Watermarkly



:-

(1) $\frac{\mu_0 I}{2R}\odot$

(2) $\frac{\mu_0 I}{2R}\left(1+\frac{1}{\pi\sqrt{2}}\right)\odot$

(3) $\frac{\mu_0 I}{2R}\left(1-\frac{1}{\pi\sqrt{2}}\right)\otimes$

(4) $\frac{\mu_0 I}{R}\left(1-\frac{1}{\pi\sqrt{2}}\right)\otimes$

5) A long solenoid has 200 turns per cm and carries a current i. The magnetic field at its centre is 6.28×10^{-2} weber/m^2. Another long solenoid has 100 turns per cm and it carries a current i/3. The value of the magnetic field at its centre is :-

(1) 1.05×10^{-4} Wb/m^2

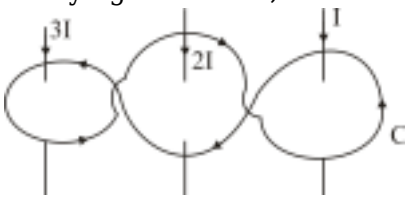
(2) 1.05×10^{-2} Wb/m^2

(3) 1.05×10^{-5} Wb/m^2

(4) 1.05×10^{-3} Wb/m^2

@examlix

6) The loop C enclosed three wires carrying current 3I, 2I and I as shown in figure then value of



$\oint \vec{B}.d\vec{\ell}$ around the closed loop is

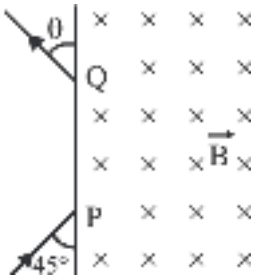
(1) $-2\mu_0 I$

(2) 0

(3) $2\mu_0 I$

(4) $6\mu_0 I$

7) A particle of mass 1.6×10^{-27} kg and charge 1.6×10^{-19} coulomb enters in a uniform magnetic field of 1 Tesla as shown in the fig. The speed of the particle is 10^7 m/s. The distance PQ will be- (θ = 45°)



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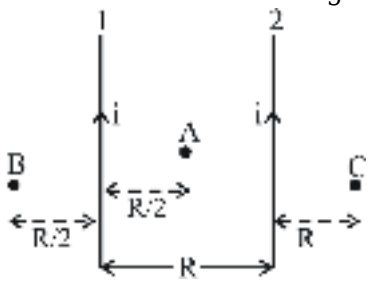
(1) 0.14 m

(2) 0.28 m

(3) 0.4 m

(4) 0.5 m

8) Two long parallel wires are at a distance R apart. They carry steady equal currents in the same directions as shown in the figure. The ratio of magnitude of magnetic fields at B and C respectively, is :-



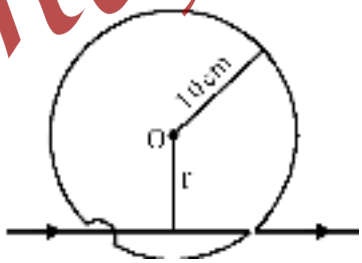
(1) 1 : 1

(2) 16 : 9

(3) 1 : 0

(4) 9 : 16

9) An infinitely long straight wire is bent as shown in figure. The circular portion has a radius of 10 cm with its center O at a distance r from the straight part. The value of r such that the magnetic field at the center O of the circular portion is zero will be



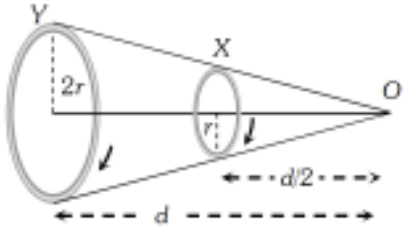
(1) $\frac{10}{\pi}$ cm

(2) $\frac{20}{\pi}$ cm

(3) $\frac{1}{5\pi}$ cm

(4) $\frac{5}{\pi}$ cm

10) Two circular coils X and Y, having equal number of turns, carry equal currents in the same sense. If the smaller coil, X is midway between O and Y, then if we represent the magnetic induction due to bigger coil Y at O as B_Y and that due to smaller coil X at O as B_X, then



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(1) $\frac{B_Y}{B_X} = 1$

(2) $\frac{B_Y}{B_X} = 2$

(3) $\frac{B_Y}{B_X} = \frac{1}{2}$

(4) $\frac{B_Y}{B_X} = \frac{1}{4}$

11) Match the column I with column II -

Column-I		**Column-II**	
(A)	Paramagnetic substance	(p)	$\chi_m = -1$
(B)	Diamagnetic substance	(q)	$\chi_m < 0$
(C)	Super conductor	(r)	$\chi_m >> 1$
(D)	Ferromagnetic substance	(s)	$\mu_r > 1$

(1) A → s, B → q, C → p, D → r

(2) A → r, B → p, C → q, D → s

(3) A → s, B → p, C → q, D → r

(4) A → r, B → s, C → p, D → q

12) A hollow cylinderical wire carries a current I, having inner and outer radii 'R' and 2R respectively. Magnetic field at a point which 3R/2 distance away from its axis is :-

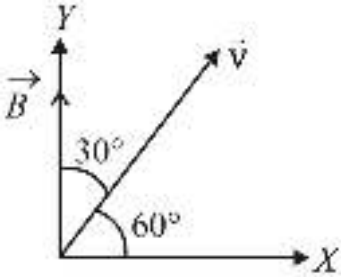
(1) $\frac{5\mu_0 I}{18\pi R}$

(2) $\frac{\mu_0 I}{36\pi R}$

(3) $\frac{5\mu_0 I}{36\pi R}$

(4) $\frac{5\mu_0 I}{9\pi R}$

13) A proton of mass 1.67×10^{-27}kg and charge 1.6×10^{-19}C is projected with a speed of 2×10^6 m/s at an angle of 60° to the X-axis as shown. If a uniform magnetic field of 0.104 Tesla is applied along Y-axis, the path of proton is



(1) A circle of radius 0.2 m and time period $\pi \times 10^{-7}$s

(2) A circle of radius 0.1 m and time period $2\pi \times 10^{-7}$s

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Watermarkly

(3) A helix of radius 0.1 m and time period $2\pi \times 10^{-7}$s

(4) A helix of radius 0.2 m and time period $4\pi \times 10^{-7}$s

14) An electron is moving along + x direction with a velocity of 6×10^6 ms^{-1}. It enters a region of uniform electric field of 300 V/cm pointing along + y direction. The magnitude and direction of the magnetic field set up in this region such that the electron keeps moving along the x direction will be:

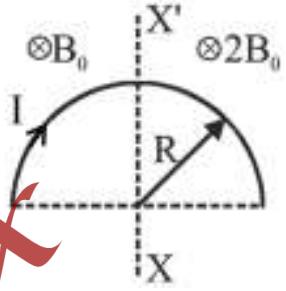
(1) 5×10^{-3} T, along +z direction

(2) 3×10^{-4} T, along –z direction

(3) 3×10^{-4} T, along +z direction

(4) 5×10^{-3} T, along –z direction

15) A semicircular wire of radius R, carrying current I is placed in a magnetic field as shown in the figure. On left side of XX', magnetic field strength is B_0 and on rigth side of XX', magnetic field



0

strength is 2B . The magnetic force experienced by the wire would be :-

(1) $3IB_0R$

(2) $2IB_0R$

(3) $\sqrt{10}\ IB_0R$

(4) $\sqrt{5}\ IB_0R$

@examlix

16) The magnetic field at a point X on the axis of a small bar magnet, is equal to the field at an equitorial point Y of the same magnet. The ratio of the distances of point X and Y from the centre of the magnet is :-

(1) 2^{-3}

(2) $2^{-1/3}$

(3) 2^3

(4) $2^{1/3}$

17) Magnetic field at the centre of a circular loop of area A is B. Then magnetic moment of the loop will be :-

(1) $\frac{BA^2}{\mu_0\pi}$

(2) $\frac{BA}{\mu_0}\sqrt{A}$

(3) $\frac{BA\sqrt{A}}{\mu_0\pi}$

Watermarkly

(4) $\frac{2BA}{\mu_0}\sqrt{\frac{A}{\pi}}$

18) A magnet of magnetic moment 25 A-m^2 is placed along '–y-axis' in a magnetic field $\vec{B} = (5\hat{i} + 3\hat{j})T$. The torque acting on the magnet is :-

(1) $-75\,\hat{j}$ Nm

(2) $125\,\hat{k}$ Nm

(3) $-125\,\hat{k}$ Nm

(4) $150\,\hat{j}$ Nm

19) Four situations given in column-I involving a magnetic dipole of dipole moment $\vec{\mu}$ placed in uniform external magnetic field $\vec{B}$. Match column-I with corresponding results in column-II.

Column-I		**Column-II**	
(A)	Magnetic dipole moment $\vec{\mu}$ is parallel to $\vec{B}$.	(p)	Force on dipole is zero
(B)	Magnetic dipole moment $\vec{\mu}$ is perpendicular to $\vec{B}$.	(q)	Torque on dipole is zero
(C)	Angle between magnetic dipole moment $\vec{\mu}$ and $\vec{B}$ is acute.	(r)	Magnitude of torque is (μB)
(D)	Angle between magnetic dipole moment $\vec{\mu}$ and $\vec{B}$ is 180°.	(s)	Potential energy of dipole due to external magnetic field is (μB)
		(t)	Magnetic dipole is in stable equilibrium

(1) A → p, q, t ; B → q, s; C → p; D → q, s

(2) A → p, s, t ; B → p, r; C → p; D → p, q, s

(3) A → p, s, t; B → q, s; C → p; D → q, s

(4) A → p, q, t; B → p, r; C → p; D → p, q, s

20) A solenoid has core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2A. If the number of turns is 1000 per metre, calculate H (magnetising field) and I (Intensity of magnetization).

(1) 2×10^3 Am^{-1}, 798×10^3 Am^{-1}

(2) 1.0 Am^{-1}, 1.5×10^5 Am^{-1}

(3) 6.8×10^{-5} Am^{-1}, 1.2×10^{-5} Am^{-1}

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(4) $2.1 \times 10^{-4}\ Am^{-1}$, $-2.6 \times 10^{5}\ Am^{-1}$

21) In half deflection method, a high resistance box is connected in series with the battery so that

(1) The deflection of the galvanometer is brought within the scale

(2) Power losses are minimized

(3) High resistance values are easily available

(4) None of the above

22) Two moving coil galvanometers, M_1 and M_2 have the following particulars:
$R_1 = 10\ \Omega$, $N_1 = 30$,
$A_1 = 3.6 \times 10^{-3}\ m^2$, $B_1 = 0.25\ T$
$R_2 = 14\ \Omega$, $N_2 = 42$,
$A_2 = 1.8 \times 10^{-3}\ m^2$, $B_2 = 0.50\ T$
(The spring constants are identical for the two meters).
Determine the ratio of the ratios of current sensitivity to voltage sensitivity of M_2 and M_1.

(1) 1.4

(2) $\frac{5}{7}$

(3) $\frac{14}{5}$

(4) 0.7

@examlix

23) To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ. If a shunt of resistance S is needed to get half deflection then G, R and S are related by the equation :

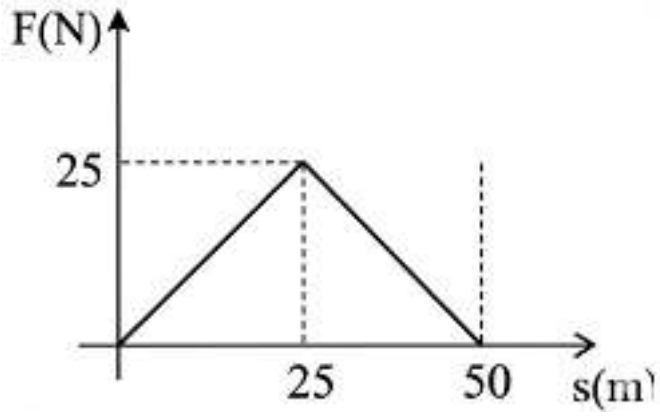
(1) 2S (R+G) = RG

(2) S (R+G) = RG

(3) 2G = S

(4) 2S = G

24) A force is applied on the body of mass 5 kg which varies with displacement of body according to the given graph. If initial speed is 10 m/s, find the final speed of the body :-



(1) 15 m/s

(2) 20 m/s

Watermarkly

(3) 10 m/s

(4) 25 m/s

25) A uniform chain of length 3 m is kept on a table 5 m high such that a length of 1 m hangs freely from the edge of the table. The total mass of the chain is 6 kg. Find the approximate speed of chain as if just touches the ground ?

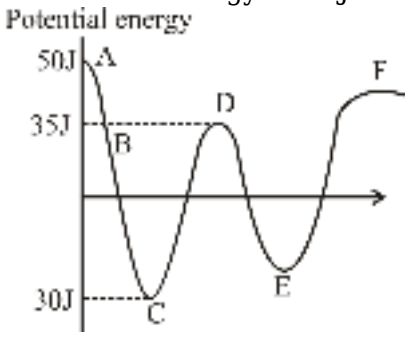
(1) 8 m/s

(2) 16 m/s

(3) 12 m/s

(4) 18 m/s

26) An object is in the region where potential energy is changing as shown in the figure. Its total mechanical energy is 30 J. If object is initially at C, then select the correct statement.



(1) Object will be able to reach at E.

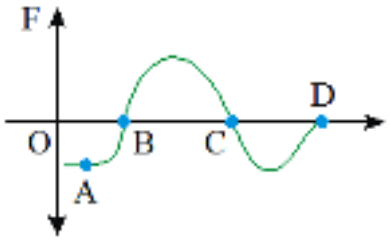
(2) At point C kinetic energy of object is 50 Joules

(3) Object will be oscillating in between B and D.

(4) Object can reach at D.

@examlix

27) A particle is being acted upon by one dimensional conservative force. In the F–x curve shown, four points A, B, C, D are marked on the curve. State which type of equilibrium does the particle have at position C.



(1) stable

(2) unstable

(3) Neutral

(4) No equilibrium

28) A force given by the relation F = 8t (N/s) acts on a body of mass 2 kg initially at rest on a smooth surface where t is in seconds. Find the work done by this force on the body during first 2 seconds of its motion. Assume direction of force remains constant and body moves in the direction of force.

(1) 64 J

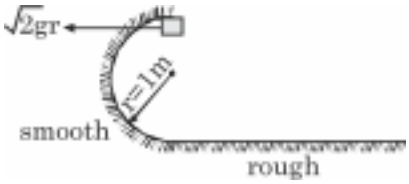
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(2) zero

(3) –64 J

(4) 32 J

29) A body is given a velocity $\sqrt{2gr}$ at the highest point of a half circular smooth track which is joined by a rough horizontal track whose co-efficient of friction is μ = 0.5. Then the distance travelled by particle before it stops on horizontal track is: (r = 1 m)



(1) 1 m

(2) 4 m

(3) 6 m

(4) none of these

30) In an experiment at the **Allen Laboratory**, a water pump is used to lift water from a ground-level tank to a storage tank located at a height of **20 meters**. The pump lifts **600 kg** of water every minute. If the efficiency of the pump is **80%**, calculate the input power required by the pump. (Take g = 10 m/s^2)

(1) 2000 W

(2) 2500 W

(3) 3000 W

(4) 1500 W

@examlix

31) Power applied to a particle varies with time as P = ($3t^2$ – 2t +1) watt, where t is in second. Find the change in its kinetic energy (in J) between time t = 2 s and t = 4 s.

(1) 46J

(2) 82J

(3) 28J

(4) 64J

32) A man with mass m jumps off of a high bridge with a bungee cord attached to his ankles. The man falls through a maximum distance H at which point the bungee cord brings him to a momentary rest before he bounces back up. The bungee cord is perfectly elastic, obeying Hooke's force law with a spring constant k, and stretches from an original length of L_0 to a maximum length L = L_0 + h. The maximum tension in the Bungee cord is four times the weight of the man. If the spring constant $k = \frac{nmg}{H}$, then find n.

(1) 1

(2) 3

(3) 8

(4) 6

Watermarkly

33) A 500 kg car moving with velocity of 36 kmh^{-1} on a straight road unidirectionally, doubles its velocity in 1 min. The average power delivered by engine for doubling the velocity is :

(1) 750 W
(2) 1050 W
(3) 1150 W
(4) 1250 W

34)

The potential energy of a 1 kg particle free to move along the x-axis is given by :-

$$U(x) = \left(\frac{x^4}{4} - \frac{x^2}{2}\right) J$$

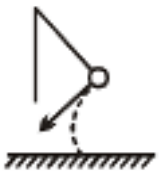
The total mechanical energy of the particle is 2 J. Then the maximum speed (in m/s) is :-

(1) $\sqrt{2}$
(2) $\frac{1}{\sqrt{2}}$
(3) 2
(4) $\frac{3}{\sqrt{2}}$

35) A girl of mass 40 kg is standing on a platform in an amusement park ride. The platform starts moving downwards with a constant acceleration of g/4. Calculate the work done by the normal reaction of the platform floor on the girl when the platform descends by a distance of 10 m. (Take g = 10 m/s^2)

(1) 1000 J
(2) 3000 J
(3) –3000 J
(4) –1000 J

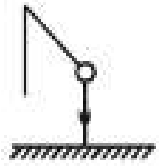
36) A pendulum bob is swinging in a vertical plane such that its angular amplitude is less than 90º. At its highest point, the string is cut. Which trajectory is possible for the bob afterwards :-

(1) 

(2) 

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(3) 

(4) 

37) Two small spheres of masses **2 kg** and **4 kg** are tied to a light string and are whirled in a horizontal circle on a frictionless table. The **2 kg** sphere is at a distance of **0.5 m** from the center, and the **4 kg** sphere is at a distance of **1.5 m** from the center. Both spheres rotate with a constant angular velocity. Find the ratio of the tension in the inner part of the string (T_1) to the tension in the outer part of the string (T_2).

(1) 1 : 6

(2) 7 : 6

(3) 6 : 7

(4) 1 : 3

38) A particle is kept fixed on a turn table rotating uniformly. As seen from the ground the particle goes in a circle, its speed is 20 cm/s and acceleration is 20 cm/s^2. The particle is now shiffted to a new position to make the radius half the original value. The new values of speed and acceleration will be:-

(1) 10 cm/s, 10 cm/s^2

(2) 10 cm/s, 80 cm/s^2

(3) 40 cm/s, 10 cm/s^2

(4) 40 cm/s, 40 cm/s^2

@examlix

39) A car is moving with uniform speed over circular bridge of radius R which subtends an angle of 90° at its centre. Find the minimum possible speed so that the car can cross the bridge without losing the contact any where :

(1) $\left(\frac{Rg}{2}\right)^{\frac{1}{2}}$

(2) $\left(\frac{Rg}{\sqrt{2}}\right)^{\frac{1}{2}}$

(3) $(Rg)^{\frac{1}{2}}$

(4) $\left(\sqrt{2}Rg\right)^{\frac{1}{2}}$

40) A grinding wheel attained a angular velocity of 20 rad/sec in 5 sec starting from rest. Find the number of revolutions made by the wheel :-

Watermarkly

(1) $\frac{\pi}{25}$ revolution

(2) $\frac{1}{\pi}$ revolution

(3) $\frac{25}{\pi}$ revolution

(4) None

41) A cyclist riding the bicycle at a speed of $14\sqrt{3}$ m/sec takes a turn around a circular road of radius $20\sqrt{3}$ m without skidding. Given g = 9.8 m/s^2. What is the banking angle of circular path ?

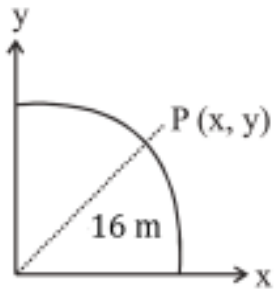
(1) 30°

(2) 90°

(3) 45°

(4) 60°

42) A point P moves in counter-clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a length $s = 6 + t^3$, where s is in meters and t is in seconds. The radius of the path is 16 m. The acceleration of 'P' when t = 2 s in nearly :-



@examlix

(1) 14 m/s^2

(2) 13 m/s^2

(3) 12 m/s^2

(4) 15 m/s^2

43) A block of mass **2 kg** slides down from rest from the top of a smooth inclined plane of height **h = 5 m**. At the bottom of the incline, it smoothly enters a vertical circular track of radius **R = 2 m**. Calculate the **Normal Reaction** force acting on the block when it reaches the lowest point of the circular track. (Take g = 10 m/s^2)

(1) 20 N

(2) 100 N

(3) 120 N

(4) 140 N

44) A car is moving in a circular horizontal track of radius 10 m with a constant speed of 10 m/sec. A plumb bob is suspended from the roof of the car by a light rigid rod of length 1.00 m. The angle made by the rod with vertical is :

(1) Zero

Watermarkly

(2) 30°

(3) 45°

(4) 60°

45) On the centre of a frictionless table a small hole is made, through which a weightless string of length 2 is inserted. On the two ends of the string two balls of the same mass m are attached. Arrangement is made in such a way that half of the string is on the table top and half is hanging below. The ball on the table top is made to move in a circular path with a constant speed v. What is the centripetal acceleration of the moving ball :

(1) mv

(2) g

(3) Zero

(4) 2mv

CHEMISTRY

1) The oxidation potential of hydrogen electrode at pH = 10, 1 atm and 25°C is :-

(1) 0.591 volt

(2) 0.059 volt

(3) –0.059 volt

(4) None

@examlix

2) The standard reduction potential values of three metallic cations, X,Y,Z are 0.52, –3.03 and –1.18 V respectively. The order of reducing power of the corresponding metals is :-

(1) Y > Z > X

(2) X > Y > Z

(3) Z > Y > X

(4) Z > X > Y

3) Given below are two statements:
Statement-I : Sodium can be obtained by the electrolysis of aqueous solution NaCl using platinum electrodes.
Statement-II : Copper metal can not reduce iron(II) ions in acidic solutions.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and II are incorrect.

(2) Both statement I and II are correct.

(3) Statement I is incorrect but statement II is correct.

(4) Statement I is correct but statement II is incorrect

4) The relationship between standard potential of cell and equilibrium constant is given by :-

Watermarkly

(1) $E^{o}_{cell} = \frac{n}{0.059} \log K_C$

(2) $E^{o}_{cell} = \frac{0.059}{n} \log K_C$

(3) $E^{o}_{cell} = \frac{\log K_C}{n}$

(4) $E^{o}_{cell} = \frac{-0.59}{n} \log K_C$

5) The molar conductance at infinte dilution for $AgNO_3$, AgCl, NaCl are 116.5, 121.6, 110.3 Ω^{-1} cm^2 mol^{-1}. Find Molar conductance of $NaNO_3$ at infinite dilution :- (in Ω^{-1} cm^2 mol^{-1})

(1) 111.4

(2) 105.2

(3) 130.6

(4) 150.2

6) The conductivity of 0.01 M solution is 10^{-4} Ω^{-1} cm^{-1}. Find its molar conductance (in terms of Ω^{-1} cm^2 mol^{-1}) :-

(1) 10

(2) 10^{-2}

(3) 10^{2}

(4) 10^{-1}

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7) For the given Nernst equation

$$E_{cell} = E^{o}_{cell} - \frac{RT}{2F} \ln \frac{[Mg^{2+}]}{[Ag^{+}]^2}$$

Which of the following representation is correct?

(1) $Ag^{+} \mid Ag \parallel Mg^{2+} \mid Mg$

(2) $Mg^{2+} \mid Mg \parallel Ag \mid Ag^{+}$

(3) $Mg \mid Mg^{2+} \parallel Ag^{+} \mid Ag$

(4) $Mg \mid Mg^{2+} \parallel Ag \mid Ag^{+}$

8) The specific conductance of a 0.1 N KCl solution at 23°C is 0.012 ohm^{-1} cm^{-1}. The resistance of cell containing the solution at the same temperature was found to be 55 ohm. The cell constant will be

(1) 0.142 cm^{-1}

(2) 0.66 cm^{-1}

(3) 0.918 cm^{-1}

(4) 1.12 cm^{-1}

9) The ionization constant of a weak electrolyte (AB) is 25×10^{-6} while the equivalent conductance of

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its 0.01 M solution is 19.6 S cm^2 eq^{-1}. The equivalent conductance of the electrolyte at infinite dilution (in S cm^2 eq^{-1}) will be

(1) 39.2

(2) 78.4

(3) 392

(4) 196

10) E° for the cell,
Zn | Zn^{2+} (aq) || Cu^{+2}(aq) | Cu is 1.10 V at 25°C. The equilibrium constant for the cell reaction

$$Zn + Cu^{2+}(aq) \rightleftharpoons Cu + Zn^{2+}(aq)$$

is of the order of

(1) 10^{-37}

(2) 10^{37}

(3) 10^{-17}

(4) 10^{17}

11) $2Ce^{4+} + Co \rightarrow 2Ce^{3+} + Co^{2+}$; $E^{o}_{Cell} = 1.89V$, $E^{o}_{Co/Co^{+2}} = 0.277V$.
Hence, $E^{o}_{Ce^{4+}/Ce^{3+}}$ is :

(1) 0.805 V

(2) 1.613 V

(3) –0.805 V

(4) –1.613 V

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12) Match List-I with List-II.

List-I (Conversion)		**List-II (Number of Faraday required)**	
(A)	1 mol of H_2O to O_2	(I)	3F
(B)	1 mol of MnO_4^- to Mn^{2+}	(II)	2F
(C)	1.5 mole of Ca from molten $CaCl_2$	(III)	1F
(D)	1 mol of FeO to Fe_2O_3	(IV)	5F

(1) A-II, B-IV, C-I, D-III

(2) A-III, B-IV, C-I, D-II

(3) A-II, B-III, C-I, D-IV

(4) A-III, B-IV, C-II, D-I

13) In a galvanic cell

(1) Chemical reaction produces electrical energy

(2) Electrical energy produces chemical reaction

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(3) Reduction occurs at anode

(4) Oxidation occurs at cathode

14) Which of the following is/ are function(s) of salt-bridge?

(1) It completes the electrical circuit with electrons flowing from one electrode to the other through external wires and a flow of ions between the two compartments through salt-bridge.

(2) It minimizes the liquid-liquid junction potential

(3) Both are correct

(4) None of the above

15) Saturated solution of KNO_3 is used to make salt-bridge because :

(1) Velocity of K^+ is greater than that of NO_3^-

(2) Velocity of NO_3^- is greater than that of K^+

(3) Velocity of both K^+ and NO_3^- are nearly the same

(4) KNO_3 is high soluble in water

16) E.M.F. of the cell
$2Ag^+ + Cu \rightarrow Cu^{+2} + 2Ag$
[Given: $E^\circ_{Ag|Ag^+}$ = –0.8 V ; $E^\circ_{Cu^{+2}|Cu}$ = 0.3 V]

(1) –0.5 V

(2) 0.5 V

(3) –1.1 V

(4) 1.1 V

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17) Given : E° (Cu^{2+}/Cu) = 0.337 V and E° (Sn^{2+} /Sn) = –0.136 V. Which of the following statements is correct ?

(1) Cu^{2+} ions can be reduced by $H_2(g)$

(2) Cu can be oxidized by H^+

(3) Sn^{2+} ions can be reduced by $H_2(g)$

(4) Cu can reduce Sn^+

18) Consider the cell potential $E^o_{Mg^{2+}|Mg}$ = –2.37 V and $E^o_{Fe^{3+}|Fe}$ = –0.04 V. The best reducing agent would be :

(1) Mg^{2+}

(2) Fe^{3+}

(3) Mg

(4) Fe

19) The number of Faradays(F) required to produce 20 g of calcium from molten $CaCl_2$ (Atomic mass

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of Ca = 40 g mol^{-1}) is :

(1) 4
(2) 1
(3) 2
(4) 3

20) When 0.1 mol MnO_4^{2-} is oxidised the quantity of electricity required to completely oxidise MnO_4^{2-} to MnO_4^- is :-

(1) 96500 C
(2) 2 × 96500 C
(3) 9650 C
(4) 96.50 C

21) Strong electrolytes are those which :

(1) dissolve readily in water
(2) conduct electricity
(3) complete dissociation into ions even at high concentration
(4) dissociate into ions at high dilution.

22) The specific conductance of a 0.01M solution of KCl is 0.0014 $ohm^{-1}cm^{-1}$ at 25°C. Its equivalent conductance ($cm^2ohm^{-1}eq^{-1}$) is :

(1) 140
(2) 14
(3) 1.4
(4) 0.14

23) The molar conductivities Λ^0_{NaOAc} and Λ^0_{HCl} at infinite dilution in water at 25°C are 91.0 and 426.2 S cm^2 mol^{-1} respectively. To calculate Λ^0_{HOAc} the additional value required is :

(1) Λ^0_{NaCl}
(2) $\Lambda^0_{H_2O}$
(3) Λ^0_{KCl}
(4) Λ^0_{NaOH}

24) The osmotic pressure of equimolar solution of $BaCl_2$, NaCl and glucose will be in the order :

(1) glucose > NaCl > $BaCl_2$
(2) $BaCl_2$ > NaCl > glucose
(3) NaCl > $BaCl_2$ > glucose
(4) NaCl > glucose > $BaCl_2$

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25) Which one of the following aqueous solutions exhibit highest boiling point?

(1) 0.01 M Na_2SO_4

(2) 0.01 M KNO_3

(3) 0.015 M urea

(4) 0.015 M glucose

26) Which of the following solutions can have boiling point less than that of both the individual components?

(1) n-Hexane and n-Heptane

(2) $CHCl_3$ and CH_3COCH_3

(3) HNO_3 and H_2O

(4) C_2H_5OH and H_2O

27) On mixing 10mL of acetone with 40mL of chloroform, the total volume of the solution is:

(1) < 50 mL

(2) > 50 mL

(3) = 50 mL

(4) cannot be predicted

28) **Statement-I:** The observed molar mass of acetic acid in benzene is more than the normal molar mass of acetic acid.
Statement-2: Molecules of acetic acid dimerise in benzene due to hydrogen bonding.

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.

(3) Statement-1 is True, Statement-2 is False.

(4) Statement-1 is False, Statement-2 is True.

29) Molarity of 720 g of pure water -

(1) 40 M

(2) 4 M

(3) 55.5 M

(4) Can't be determined

30) Osmotic pressure of 30% solution of glucose is 1.20 atm and that of 3.42% solution of cane-sugar is 2.5 atm. The osmotic pressure of the mixture containing equal volumes of the two solutions will be :

(1) 2.5 atm

(2) 3.7 atm

(3) 1.85 atm

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(4) 1.3 atm

31) Which of the following is independent of temperature :

(1) Molarity
(2) % W/V
(3) Molality
(4) Both (1) and (2)

32) Molarity represents number of moles of solute present in :

(1) One litre solvent
(2) One kg of solvent
(3) One litre of solution
(4) None of these

33) Which of the following is not a colligative property ?

(1) Elevation in boiling point
(2) Osmotic pressure
(3) Depression in freezing point
(4) Boiling point

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34) If 0.1 M solution of glucose and 0.1 M solution of urea are placed on two sides of the semi-membrane of equal heights, then it will be correct to say that:

(1) there will be no net movement across the membrane
(2) glucose will flow towards urea solution
(3) urea will flow towards glucose solution
(4) water will flow from the urea solution to glucose solution

35) **Assertion (A)**: 0.1 M solution of NaCl has greater osmotic pressure. than 0.1 M solution of glucose at same temperature.
Reason (R): In solution, NaCl dissociates to produce more number of particles.

(1) If both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) If both (A) and (R) are correct and (R) is not the correct explanation of (A).
(3) If (A) is correct but (R) is incorrect.
(4) If (A) is incorrect but (R) is correct.

36) The amount of ice that will separate out on cooling a solution containing 50 g ethylene glycol (molar mass = 62 g/mol) in 200 g water to –9.3° C is : (K_f = 1.86 K molality^{-1})

(1) 38.71 g
(2) 38.71 mg
(3) 42 g

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(4) 42 mg

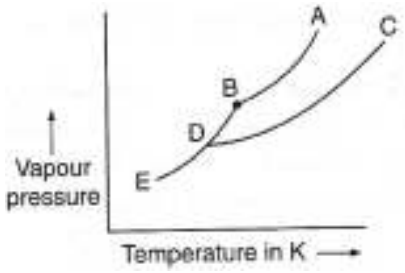
37) Which one of the following statement is false?

(1) Raoult's law states that the vapour pressure of a component over a binary solution of volatile liquids is directly proportional to its mole fraction

(2) Two sucrose solutions of the same molality prepared in different solvents will have the same depression in freezing point.

(3) The correct order of osmotic pressures of 0.01 M solution of each compound is: $BaCl_2$ > KCl > glucose

(4) None of these



38) Which of the following is correctly matched ?

(1) Curve (A-B) - solution

(2) Curve (C-D) - liquid solvent

(3) Curve (B-D-E) - frozen solvent

(4) Curve (A-B) - frozen solution

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39) 18 g glucose and 34.2 g sucrose are dissolved in water to make 2 litre solution at 27°C. Osmotic pressure of this solution will be :

(1) 3.46 atm

(2) 2.46 atm

(3) 4.46 atm

(4) 1.46 atm

40) A solution containing 7 g of a solute (molar mass 210 g mol^{-1}) in 350 g of acetone raised the boiling point of acetone from 56°C to 56.3°C. The value of ebullioscopic constant of acetone in K kg mol^{-1} is:

(1) 2.66

(2) 3.15

(3) 4.12

(4) 2.86

41) The heat of neutratization of any strong acid and a strong base is always constant. This is because :

(1) Both the acid and base undergo complete ionization

(2) During the neutralization, salt and water are formed

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(3) 1 mole of water is formed from H^+ and OH^- ions

(4) The reaction is exothermic

42) The equivalent weight of sodium carbonate [Na_2CO_3] is

(1) 40

(2) 53

(3) 55.85

(4) 63

43) The number of grams of solute per 100 ml of solution is known as

(1) Normality

(2) Molarity

(3) % weight by volume

(4) Mole fraction

44) Match the following

1	Strong acid –Strong base Titration	P	Not performedpractically
2	Strong base –Weak acid titration	Q	NH_4OH Vs HCl
3	Strong acid –Weak base titration	R	NaOH Vs HCl
4	Weak acid –Weak base titration	S	NaOH Vs CH_3COOH

(1) 1-R, 2-P, 3-S, 4-Q

(2) 1-R, 2-S, 3-P, 4-Q

(3) 1-R, 2-S, 3-Q, 4-P

(4) 1-S, 2-Q, 3-R, 4-P

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45) Which of the following is not a lyophilic colloid?

(1) Milk

(2) Gum

(3) Fog

(4) Blood

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1) Histones are enriched with :

(1) Lysine & Arginine

(2) Leucine & Lysine

(3) Phenylalanine & Leucine

(4) Phenylalanine & Arginine

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2) Who proposed that the genetic code for amino acids should be made up of three nucleotides ?

(1) George Gamow
(2) Francis Crick
(3) Jacque Monod
(4) Franklin Stahl

3) Which factor is important for termination of transcription ?

(1) α (alpha)
(2) σ (sigma)
(3) ρ (rho)
(4) γ (gamma)

4) The length of DNA having 23 base pairs :-

(1) 78 Å
(2) 78.4 Å
(3) 78.2 Å
(4) 74.8 Å

5) In Hershey & Chase experiment radioactive S^{35} detected in :-

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(1) Supernatant
(2) Pellet
(3) Both
(4) None

6) In a nucleoside unit N-glycosidic bond is formed between :-

(1) Nitrogenous base and sugar
(2) Phosphoric acid and sugar
(3) Sugar and sugar
(4) Two nucleotide unit

7) Which of the following enzyme did not affect transformation in Griffith's experiment :-

(1) Proteases
(2) RNases
(3) DNase
(4) (1) and (2) Both

8) **Assertion:** The pitch of the DNA helix is 3.4 nm
Reason: The distance between 2 base pairs is 0.34 nm & there are 10 bp in each turn.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

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(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

9) What is the major hallmark of double helix model of DNA given by Watson and Crick ?

(1) Base pairing between two strands
(2) Presence of phosphodiester bond
(3) Acidic nature of DNA
(4) Presence of pentose sugar

10) Which of the following feature generates approximately uniform distance between the two strands of DNA ?

(1) Presence of hydrogen bond
(2) Antiparallel polarity of two strands
(3) One purine always comes opposite to a pyrimidine
(4) Ester bond

11) Genetic material is changed in an organism according to :-

(1) Different stages of life cycle
(2) Age
(3) Change in physiology of organism
(4) None of the above

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12) How many nitrogen base pairs are there in the DNA of *E.coli ?*

(1) 5386 bp
(2) 48502 bp
(3) 4.6×10^6 bp
(4) 3.3×10^9 bp

13) A cytidine is formed through N-glycosidic linkage by joining :-

(1) Phosphate group with adenine
(2) Cytosine with cytosine
(3) Ribose sugar with cytosine
(4) Deoxyribose sugar with phosphate group

14) A double stranded linear DNA molecule has total 1000 nucleotides. Calculate the number of phosphodiester bonds in it.

(1) 1000
(2) 999
(3) 998

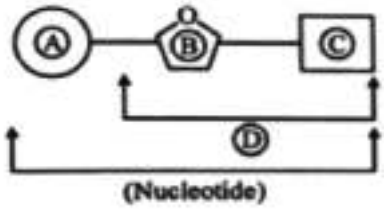
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(4) 500

15) DNA was extracted from a bacterium. The proportion of thymine was found to be 15% then calculate the amount of guanine.

(1) 15%
(2) 70%
(3) 35%
(4) 30%

16) Identify the missing words A, B, C and D in the following figure related to DNA structure :



	A	B	C	D
(1)	Phosphate	Deoxyribose	Purine	Nucleoside
(2)	Nucleoside	Phosphate	Deoxyribose	Purine
(3)	Phosphate	Purine	Deoxyribose	Nucleoside
(4)	Deoxyribose	Nucleoside	Purine	Phosphate

(1) 1
(2) 2
(3) 3
(4) 4

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17) Which structure observed as bead on string during packaging of DNA ?

(1) Nucleotide
(2) Nucleoside
(3) Nucleosome
(4) Histone

18) The DNA polymerase on their own cannot initiate the process of replication so _____ is required to initiate the replication process and it is synthesised by _____,

(1) DNA primer, DNA polymerase
(2) RNA primer, RNA polymerase like enzyme
(3) DNA primer, Primase
(4) RNA primer, reverse transcriptase

19) Match the Column I with Column II :-

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Column-I		Column-II	
(I)	Taylor	(a)	N^{15}
(II)	Hershey and Chase	(b)	*Streptococcus pneumoniae*
(III)	Meselson and Stahl	(c)	*Vicia faba*
(IV)	Griffth	(d)	Bacteriophage

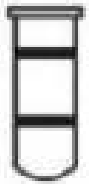
(1) I-c, II-a, III-d, IV-b

(2) I-c, II-b, III-d, IV-a

(3) I-c, II-d, III-a, IV-b

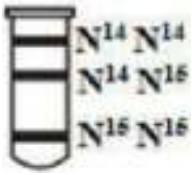
(4) I-c, II-a, III-b, IV-d

20) Light DNA molecule is replicated in a heavy medium. After the four successive replication of DNA, 16 DNA molecules are formed. Which one the following is true for these DNA.

(1) 

(2) 

(3) 

(4) 

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21) During replication of DNA, Okazaki fragments are fromed in the direction of:

(1) 3' → 5'

(2) 5'→3'

(3) 5'→ 5'

(4) 3' → 3'

22) A molecular biologist analysed a nucleic acid sample and found that it contained the N_2 bases in following proportion A = 20%, T = 15%, C = 35%, G = 30% this data indicates that :-

(1) Nucleic acid in the sample is single stranded DNA

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(2) Nucleic acid in the sample is double stranded DNA
(3) Nucleic acid in the sample is double stranded RNA
(4) Nucleic acid in the sample is single stranded RNA

23) From the following, identify the correct combination of salient features of Genetic Code

(1) Universal, Non-ambiguous, Overlapping
(2) Degenerate, Overlapping, Commaless
(3) Universal, Ambiguous, Degenerate
(4) Degenerate, Non-overlapping, Non- ambiguous

24) Which of following DNA strand required many RNA primer at the time of DNA replication :-

(1) Leading strand
(2) Lagging strand
(3) Both (1) and (2)
(4) Continuous-strand

25) The essential life processes such as metabolism, translation and splicing evolved around :-

(1) DNA
(2) Protein
(3) RNA
(4) Lipid

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26) Find out the correct statements from the following :
(a) In lac-operon structural genes are regulated by a common promoter and regulatory gene
(b) In lac-operon one regulatory gene (the i gene) is present, here the term i refers to the inducer
(c) Lactose is the substrate for the enzyme betagalactosidase and it regulates switching on and off the operon
(d) The y-gene of lac-operon codes for permease
(e) The z-gene codes for beta-galactosidase

(1) a and c only
(2) a, b and e only
(3) a, c and e only
(4) a, c, d and e

27) **Assertion :** DNA fingerprinting is mainly based on polymorphism in VNTRs (Variable Number of Tandem Repeats).
Reason : VNTRs are highly conserved regions of DNA that do not show variation among individuals.

(1) Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion & Reason are False.

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28) DNA is a long polymer of nucleotides. If in a ds DNA, 4000 base pairs are found then what will be the number of nucleotides, sugar molecules, phosphate molecules?

(1) 8000, 4000, 4000

(2) 4000, 4000, 8000

(3) 4000, 8000, 4000

(4) 8000, 8000, 8000

29) Match the following with respect to Transcription in Prokaryotes:

Column I		Column II	
1.	σ (Sigma) factor	(a)	Helps in promoter recognition and initiation
2.	Template strand	(b)	Serves as actual template for RNA synthesis
3.	Coding strand	(c)	Same sequence as RNA (except U→T)
4.	RNA Polymerase	(d)	Catalyzes formation of RNA

(1) 1-a, 2-b, 3-c, 4-d,

(2) 1-d, 2-b, 3-c, 4-a,

(3) 1-c, 2-d, 3-a, 4-b,

(4) 1-c, 2-d, 3-b, 4-a,

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30) A DNA sequence is given: 5'-ATG AAG TCG TAA-3' After mutation, one nucleotide "A" is added after the first codon (ATG). What will be the result of this mutation?

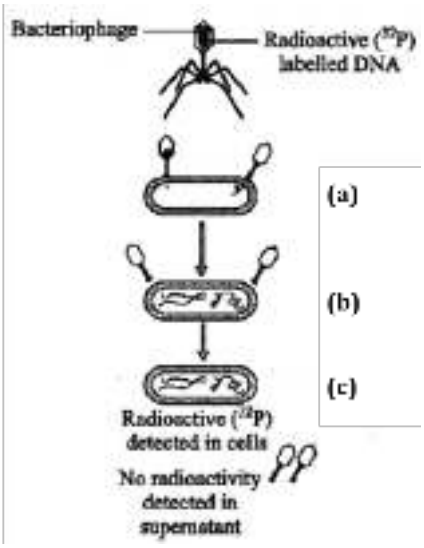
(1) Only one amino acid will change

(2) No effect due to degeneracy of code

(3) Frameshift mutation leading to completely altered protein sequence

(4) Formation of identical protein

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31) Recognised the figure and find out the correct matching.

(1) a = blending; b = centrifugation; c = infection
(2) c = blending; a = centrifugation; b = infection
(3) b = blending; c = centrifugation; a = infection
(4) c = blending; b = centrifugation; a = infection

32) If Griffith experiment, mice injected with mixture of heat killed S-type bacteria and living R-type bacteria were killed because:

(1) Living R-type bacteria changed into S-type
(2) Heat killed S-type bacteria were rejuvenated
(3) Both (1) and (2)
(4) None of these

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33) Chemically, RNA is __i__ reactive and __ii__ stable as compared to DNA.

(1) i = equally; ii = equally
(2) i = less; ii = more
(3) i = more; ii = less
(4) i = more; ii = equally

34) Choose the correct option w.r.t. RNA :

(1) presence of thymine in place of uracil
(2) Absence of free 2'OH in sugar
(3) Mutates at faster rate
(4) Is non catalytic

35) The packaging of chromatin in eukaryotes at higher level requires set of proteins that are collectively referred to as.

(1) Histones
(2) Polyamines

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(3) Non Histone chromosomal protein

(4) Globins

36) If the length of bacterial DNA is 6.80 mm, then how many base pairs are present in this bacteria.

(1) 4.0×10^7

(2) 2.0×10^7

(3) 4.6×10^6

(4) 3.3×10^9

37) Complete the following statement by filling gaps (a) and (b) :
Euchromatin is said to be transcriptionally __a__ chromatin, whereas heterochromatin is __b__.

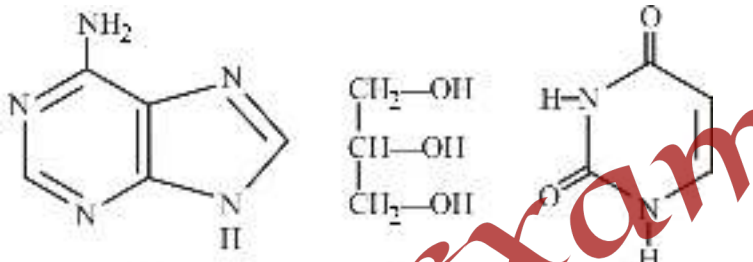
(1) a - active; b- active

(2) a - active; b - inactive

(3) a - inactive; b - inactive

(4) a - active; b - highly active

38)



(A) (B) (C)

Which of the above may be present in both RNA & DNA ?

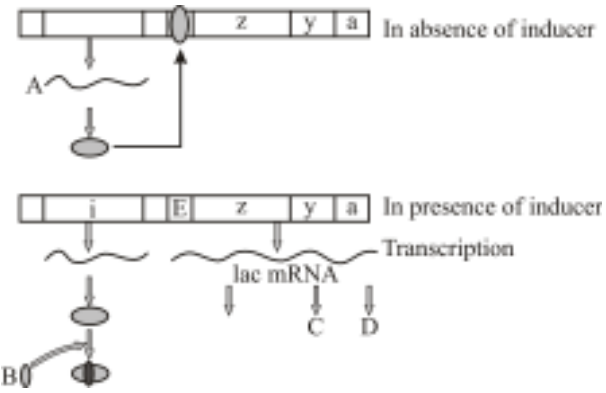
(1) A

(2) B

(3) C

(4) Both A & B

39)



Identify A, B, C, D, E :

(1) A = inducer, B = repressor, C = inhibitor, D = mRNA, E = operator

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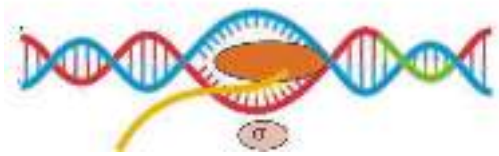
A = repressor mRNA, B = inducer,
(2) C = permease, D = transacetylase,
E = operator

A = repressor, B = operator,
(3) C = β-galactosidase, D = transacetylase,
E = promoter

A = operator, B = inducer,
(4) C = permease, D = RNA polymerase,
E = lac mRNA

40) Find out the missing figure with correct option in the given figure.





(1) 

(2) 

(3) 

(4) 

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41) In the given picture correctly identify A, B and C :-

	A	B	C
(1)	Sigma factor	Rho factor	Stop codon
(2)	RNA polymerase	Sigma factor	Stop codon

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(3)	Rho factor	RNA polymerase	Terminator
(4)	RNA polymerase	Sigma factor	Rho factor

(1) 1

(2) 2

(3) 3

(4) 4

42) In a m-RNA sequence of N_2-base is 5' AUG GUG CUC AAA 3'. What is the correct sequences of anticodons which recognize codons of m-RNA :-



UUU GAG UAC CAC **(a)** **(b)** **(c)** **(d)**

(1) a, b, c, d

(2) d, a, b, c

(3) c, d, b, a

(4) d, c, b, a

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43) **Assertion (A) :** Genetic code is nearly universal.
Reason (R) : It is same for all organism except in mitochondria & some protozoans.
Choose the correct answer:-

(1) (A) is correct while (R) is incorrect

(2) (A) is incorrect while (R) is correct

(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

44) **Assertion (A) :** Mutation results in changes in genotype and phenotype of an organism.
Reason (R) : Mutation leads to alteration of DNA sequences that is variations in DNA.
Choose the correct answer from the options given below.

(1) Both (A) and (R) are correct and (R) is correct explanation of (A).

(2) Both (A) and (R) are correct and (R) is not correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

45) **Assertion :** The UTRs are present at both 5' -end (before start codon) and at 3' -end (after stop codon) of mRNA.
Reason : UTRs are required for efficient transcription process.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

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(4) Both Assertion & Reason are False.

46) Which of the following traits of the garden pea plant was NOT studied by Gregor Mendel?

(1) Seed shape
(2) Pod colour
(3) Leaf size
(4) Flower position

47) In a monohybrid cross of tall and dwarf pea plants, what proportion of the F_2 generation is dwarf?

(1) 1/2
(2) 1/4
(3) 3/4
(4) 1/3

48) What is the genotypic ratio of the F_2 generation in a case of incomplete dominance, such as flower colour in Snapdragon?

(1) 3 : 1
(2) 9 : 3 : 3 : 1
(3) 1 : 2 : 1
(4) 1 : 1

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49) Which type of sex determination mechanism is observed in grasshoppers?

(1) XY type
(2) XO type
(3) ZW type
(4) ZZ type

50) Which amino acid is substituted by Valine in the beta globin chain to cause sickle-cell anaemia?

(1) Tyrosine
(2) Phenylalanine
(3) Glutamic acid
(4) Histidine

51) **Assertion (A):** The cross between a tall plant from F_2 with a dwarf plant is called a test cross.
Reason (R): A test cross is used to determine the genotype of an organism showing a dominant phenotype.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.

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(4) Both (A) and (R) are false.

52) **Assertion (A):** The ABO blood grouping in human beings is a good example of co-dominance.
Reason (R): When I^A and I^B alleles are present together, they both express their own types of sugars.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false.

53) **Assertion (A):** Genes located on the same chromosome always assort independently.
Reason (R): Morgan found that when two genes are situated on the same chromosome, the proportion of parental gene combinations is much higher than the non-parental type.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

54) **Assertion (A):** Phenylketonuria is an example of a pleiotropic gene.
Reason (R): The disease manifests itself through multiple phenotypic expressions like mental retardation and reduction in hair and skin pigmentation.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false.

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55) **Assertion (A):** In human beings, the sex of the child is determined by the genetic makeup of the sperm.
Reason (R): Females produce only one type of ovum with an X-chromosome, whereas males produce 50% sperms with X-chromosome and 50% with Y-chromosome.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false.

56) **Statement I:** Thalassemia is a quantitative problem of synthesising too few globin molecules.
Statement II: Sickle-cell anaemia is a qualitative problem of synthesising an incorrectly functioning globin.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Only Statement I is correct
(4) Only Statement II is correct

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57) **Statement I:** Down's syndrome results in the gain of an extra copy of chromosome 21.
Statement II: Turner's syndrome results due to the gain of an extra X chromosome in human females.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Only Statement I is correct
(4) Only Statement II is correct

58) **Statement I:** In birds, the total number of chromosomes is the same in both males and females.
Statement II: Birds exhibit male heterogamety and produce two different types of gametes.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Only Statement I is correct
(4) Only Statement II is correct

59) **Statement I:** Mendel's work on the inheritance of characters was published in 1865 but remained unrecognised till 1900.
Statement II: In 1900, de Vries, Correns, and von Tschermak independently rediscovered Mendel's results.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Only Statement I is correct
(4) Only Statement II is correct

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60) **Statement I:** In polygenic inheritance, the phenotype reflects the contribution of each allele, meaning the effect of each allele is additive.
Statement II: Human skin colour is a classic example of polygenic inheritance controlled by three genes.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Only Statement I is correct
(4) Only Statement II is correct

61) Match the genetic disorders in Column I with their characteristics in Column II.

Column I		Column II	
A.	Down's syndrome	i.	45 with XO
B.	Klinefelter's syndrome	ii.	Substitution of Glutamic acid by Valine
C.	Turner's syndrome	iii.	Trisomy of chromosome 21

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D	Sickle-cell anaemia	iv.	Karyotype of 47, XXY

(1) A-iii, B-iv, C-i, D-ii

(2) A-iv, B-iii, C-ii, D-i

(3) A-iii, B-i, C-iv, D-ii

(4) A-ii, B-iv, C-i, D-iii

62) Match the traits studied by Mendel in pea plants (Column I) with their recessive forms (Column II).

Column I	Column II
A. Seed shape	i. Terminal
B. Flower position	ii. Green
C. Seed colour	iii. Constricted
D. Pod shape	iv. Wrinkled

(1) A-iv, B-i, C-ii, D-iii

(2) A-iii, B-iv, C-i, D-ii

(3) A-iv, B-ii, C-i, D-iii

(4) A-ii, B-i, C-iv, D-iii

63) Match the Scientists in Column I with their respective contributions in Column II.

Column I	Column II
A. T.H. Morgan	i. X body
B. Sutton and Boveri	ii. Genetic maps
C. Henking	iii. Chromosomal theory of inheritance
D. Alfred Sturtevant	iv. Linkage

(1) A-iv, B-iii, C-i, D-ii

(2) A-iii, B-iv, C-ii, D-i

(3) A-iv, B-i, C-iii, D-ii

(4) A-ii, B-iii, C-iv, D-i

64) Match the genotype of the human blood group (Column I) with the phenotype (Column II).

Column I	Column II
A. I^Ai	i. Blood type O
B. I^AI^B	ii. Blood type B
C. I^Bi	iii. Blood type AB
D. ii	iv. Blood type A

(1) A-iv, B-iii, C-ii, D-i

(2) A-iii, B-iv, C-i, D-ii

(3) A-iv, B-ii, C-iii, D-i

(4) A-i, B-ii, C-iii, D-iv

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65) Match the terms related to inheritance (Column I) with their descriptions (Column II).

Column I	Column II
A. Homozygous	i. Contains alleles expressing contrasting traits
B. Heterozygous	ii. First progeny of a cross between two parents
C. Phenotype	iii. Identical alleles for a trait
D. F_1	iv. The observable descriptive character

(1) A-iii, B-i, C-iv, D-ii

(2) A-i, B-iii, C-ii, D-iv

(3) A-iii, B-ii, C-i, D-iv

(4) A-iv, B-i, C-iii, D-ii

66) In a dihybrid cross between a plant with round yellow seeds (RRYY) and wrinkled green seeds (rryy), what is the expected frequency of plants with genotype Rryy in the F_2 generation?

(1) 1/16

(2) 2/16

(3) 3/16

(4) 4/16

67) Morgan used *Drosophila melanogaster* for his genetic experiments. Which of the following is NOT a reason for its suitability?

(1) They can be grown on simple synthetic medium.

(2) They complete their life cycle in about two weeks.

(3) Male and female flies are indistinguishable.

(4) A single mating produces a large number of progeny flies.

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68) In the pedigree analysis, what does a diamond symbol represent?

(1) Mating between relatives

(2) Sex unspecified

(3) Affected individual

(4) Unaffected male

69) Which of the following is a symptom of Down's syndrome?

(1) Rudimentary ovaries

(2) Gynaecomastia

(3) Furrowed tongue and partially open mouth

(4) Elongated sickle like structure of RBC

70) The mechanism of sex determination in honey bees is called:

(1) Male heterogamety

(2) Female heterogamety

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(3) Haplodiploid sex-determination system
(4) XO type sex-determination

71) Which of the following conditions correctly describes Turner's syndrome?

(1) Trisomy 21
(2) 47, XXY
(3) 45 with XO
(4) Polyploidy

72) In which organism did Henking (1891) first trace the 'X body' during spermatogenesis?

(1) Human
(2) Bird
(3) Pea plant
(4) Insects

73) A disease controlled by a single gene exhibiting multiple phenotypic expressions, such as mental retardation and reduction in hair pigmentation, is called:

(1) Polygenic trait
(2) Pleiotropy
(3) Co-dominance
(4) Incomplete dominance

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74) Starch synthesis in pea seeds is controlled by one gene. Heterozygotes (Bb) produce round seeds with intermediate-sized starch grains. If starch grain size is considered the phenotype, what does it illustrate?

(1) Complete dominance
(2) Co-dominance
(3) Incomplete dominance
(4) Multiple alleles

75) If a colour blind man marries a woman who is homozygous for normal colour vision, what is the probability of their son being colour blind?

(1) 0%
(2) 25%
(3) 50%
(4) 100%

76) The frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes was used by:

(1) T.H. Morgan
(2) Sutton and Boveri

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(3) Alfred Sturtevant

(4) Gregor Mendel

77) A child has blood group O. If the father has blood group A and the mother has blood group B, what are the possible genotypes of the parents?

(1) Father $I^A I^A$, Mother $I^B I^B$

(2) Father I^A i, Mother I^B i

(3) Father I^A i, Mother $I^B I^B$

(4) Father $I^A I^A$, Mother I^B i

78) Which of the following laws of Mendel is based on the fact that the alleles do not show any blending and that both the characters are recovered as such in the F_2 generation?

(1) Law of Dominance

(2) Law of Segregation

(3) Law of Independent Assortment

(4) Chromosomal Theory of Inheritance

79) Experimental verification of the chromosomal theory of inheritance was provided by:

(1) Sutton and Boveri

(2) T.H. Morgan and his colleagues

(3) Gregor Mendel

(4) Hugo de Vries

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80) A diploid organism is heterozygous for 4 loci. How many types of gametes can be produced?

(1) 4

(2) 8

(3) 16

(4) 32

81)

In Phenylketonuria, the affected individual lacks an enzyme that converts the amino acid phenylalanine into :

(1) Tyrosine

(2) Valine

(3) Glutamic acid

(4) Tryptophan

82) Which of the following describes the genotype and phenotype of Klinefelter's syndrome?

(1) 45 with XO, sterile female

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(2) 47 with XXY, sterile male with some feminine development
(3) Trisomy 21, mentally retarded male or female
(4) 47 with XYY, highly aggressive male

83) Haemophilia is categorized as which type of genetic disorder?

(1) Autosomal dominant trait
(2) Autosomal recessive trait
(3) Sex-linked (X-linked) recessive trait
(4) Sex-linked dominant trait

84) What is the cross between an F_1 individual and its homozygous recessive parent called?

(1) Out cross
(2) Test cross
(3) Monohybrid cross
(4) Dihybrid cross

85) In Morgan's dihybrid cross with *Drosophila*, the genes for white eye and miniature wing showed what percentage of recombination?

(1) 1.3%
(2) 37.2%
(3) 62.8%
(4) 98.7%

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86) A woman whose father was colour blind marries a normal man. What is the probability that their sons will be colour blind?

(1) 0%
(2) 25%
(3) 50%
(4) 100%

87) An alteration in a single base pair of DNA is known as:

(1) Frameshift mutation
(2) Point mutation
(3) Chromosomal aberration
(4) Polyploidy

88) Out of the three possible genotypes for the sickle-cell trait (Hb^A Hb^A, Hb^A Hb^s, Hb^s Hb^s), which one shows the diseased phenotype?

(1) Hb^A Hb^A
(2) Hb^A Hb^S

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(3) Hb^S Hb^S

(4) Both Hb^A Hb^S and Hb^S Hb^S

89) In birds, the sex of the offspring is strictly determined by:

(1) The type of sperm that fertilizes the egg.

(2) The temperature of incubation.

(3) The type of egg that is fertilized.

(4) The number of autosomes present.

90) Which of the following traits in humans is controlled by polygenic inheritance, wherein the trait is influenced by three or more genes?

(1) ABO Blood grouping

(2) Sickle cell anaemia

(3) Phenylketonuria

(4) Human skin colour

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NOVA

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	4	2	2	1	1	2	1	3	1	3	3	1	3	4	4	2	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	2	1	1	3	1	1	3	2	1	3	4	4	3	3	2	1	2	3
Q.	41	42	43	44	45															
A.	4	4	3	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	3	2	2	1	3	2	3	2	2	1	1	3	3	2	1	3	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	1	1	2	1	4	1	1	3	3	3	3	4	1	1	1	2	3	2	2
Q.	86	87	88	89	90															
A.	1	2	3	3	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	3	3	1	1	4	1	1	3	4	3	3	3	3	1	3	2	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	4	2	3	4	3	4	1	3	3	1	3	3	3	2	2	1	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	4	1	3	3	2	3	2	3	1	1	4	1	1	1	3	3	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	1	1	1	1	2	3	2	3	3	3	4	2	3	1	3	2	2	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	2	3	2	2	3	2	3	3	4										

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SOLUTIONS

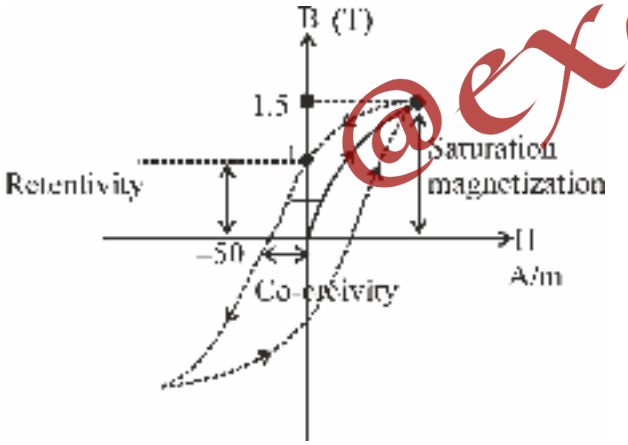
PHYSICS

1) **Explaining :** To find the correct property of magnetic force on moving charge.
Concept : This question is based on $\vec{F}_M \perp \vec{v}$
$F_M \perp \vec{B}$
Solution : Magnetic force $\perp$ moving charge
$\Rightarrow$ No work done $\Rightarrow$ No change in speed.

Electric force may change dir. or value or both. $\Rightarrow \vec{a} = \frac{q\vec{E}}{m}$
Final Answer : (2)

2)

Question Explanation:
The question asks you to identify three properties—retentivity, coercivity, and saturation—from a B-H (magnetic induction vs. magnetizing field) curve for a ferromagnetic material.
Concept: B-H Hysteresis Loop
Solution:



Retentivity = 1.0 T
Co-ercivity = 50 A/m
Saturation = 1.5 T
Final answer : Option (2)
1.0 T, 50 A/m and 1.5 T

3)

Explaining the question :
1. Explanation of Question : Determine the kinetic energy required for an alpha particle to travers a circular path of radius R in the same uniform magnetic field where a proton with 1 MeV kinetic energy.

Concept : motion of charged particle in magnetic field.

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Formula :

The kinetic energy (KE) of a particle is $KE = \frac{1}{2}mv^2$ solving for v → $v = \sqrt{\frac{2KE}{m}}$ substituting into the radius formula $R = \frac{m\sqrt{\frac{2KE}{m}}}{qB} = \frac{\sqrt{2mKE}}{qB}$ rearranging for kinetic energy $KE = \frac{(eBR)^2}{2m_p} = \frac{\sqrt{2m.KE}}{qB}$ Rearranging for kinetic energy:

$KE = \frac{(qBR)^2}{2m}$

Solution :

For the proton $KE = \frac{(eBR)^2}{2m_p}$ given $KE_p = 1MeV$ we have 1 $MeV = \frac{(eBR)^2}{2m_p}$,

For the alpha particle: $KE_a = \frac{(2eBR)^2}{2 \times 4m_p} = \frac{4(eBR)^2}{8m_p} = \frac{(eBR)^2}{2m_p}$ therefore : KE_a =1MeV

Final Answer :

Hence, option (4) is correct.

4) **Explaining :** To find magnetic file at the centre of the circular loop.

Concept : This question is based on magnetic field.

Solution : Magnetic field due to 'AB';

$\vec{B}_1 = \frac{\mu_0 I}{4\pi R}\left[\sin\frac{\pi}{2} - \sin\frac{\pi}{4}\right]\otimes = \frac{\mu_0 I}{4\pi R}\left[1 - \frac{1}{\sqrt{2}}\right]\otimes$

Magnetic field due to circular loop,

$\vec{B}_2 = \frac{\mu_0 I}{2R}\odot$

Magnetic field due to straight wire BC,

$\vec{B}_3 = \frac{\mu_0 I}{4\pi R}\left[\sin\frac{\pi}{2} + \sin\frac{\pi}{4}\right]\odot = \frac{\mu_0 I}{4\pi R}\left[1 + \frac{1}{\sqrt{2}}\right]\odot$

$\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$

$= \left(\frac{\mu_0 I}{2R} + \frac{2\mu_0 I}{4\pi R}\frac{1}{\sqrt{2}}\right)\odot = \frac{\mu_0 I}{2R}\left[1 + \frac{1}{\sqrt{2}\pi}\right]\odot$

Final Answer : (2)

5) **Explaining :** To find magnetic field at given point.

Concept : This question is based on $B = \mu_0 ni$.

Solution : $B = \mu_0\, nI$

$\frac{B_2}{B_1} = \frac{n_2 I_2}{n_1 I_1}$

$\frac{B_2}{6.28 \times 10^{-2}} = \frac{100}{200} \times \frac{i/3}{i}$

$B_2 = 1.05 \times 10^{-2}\ \frac{wb}{m^2}$

Final Answer : (2)

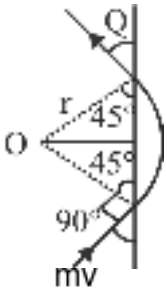
6) **Explaining :** To evaluate the integral $\oint \vec{B}.d\vec{\ell}$.

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Concept : This question is based on Ampere's Circutal Law.
Solution :
$\oint \vec{B}.d\vec{\ell} = \mu^{0}(I_{end})$
Here I_{end} = –I + 2I – 3I = –2I
Final Answer : (1)

7) **Explaining :** To find the distance PQ.
Concept : This question is based on motion in limited magnetic field.
Solution : $\frac{PQ}{2} = r\cos 45°$
PQ = 2r cos 45°



$r = \frac{mv}{qB}$
$PQ = 2r\left(\frac{1}{\sqrt{2}}\right)$
$PQ = \frac{\sqrt{2}mv}{qB}$
$PQ = \frac{\sqrt{2}\left(1.6\times10^{-27}\right)(10^{7})}{\left(1.6\times10^{-19}\right)(1)}$
PQ = 0.14m
Final Answer : (1)

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8) **Explaining :** To find ratio of mag. field at given points.
Concept : This question is based on $B_{st.\ wire\ long} = \frac{\mu_0 I}{2\pi r_{\perp}}$.
Solution :
$B_B = \frac{\mu_0 I}{2\pi\frac{R}{2}} + \frac{\mu_0 I}{2\pi\frac{3R}{2}} = \frac{4\mu_0 I}{3\pi R}$
$B_C = \frac{\mu_0 I}{2\pi R} + \frac{\mu_0 I}{2\pi(2R)} = \frac{3\mu_0 I}{4\pi R}$
$\frac{B_B}{B_C}$ = 16: 9
Final Answer : (2)

9) **Explaining :** To find distance r so that magnetic field at centre is zero as given.
Concept : This question is based on $B_{circular\ loop} = \frac{\mu_0 I}{2R}$.
Solution : $B_{circular\ loop} = -B_{wire} \Rightarrow \frac{\mu_0 I}{2\times10} = \frac{\mu_0 I}{2\pi r}$

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$r = \frac{10}{\pi}$ cm

Final Answer : (1)

10) **Explaining :** To find the ratio of magnetic fields as given cond.^{n}

Concept : This question is based on Mag. field on axis of a circular current carriers coil :

$$B = \frac{\mu_0 I R^2}{2\left(R^2 + x^2\right)^{3/2}}$$

Solution :

Magnetic field at O due to bigger coil Y, is

$$B_Y = \frac{\mu_0 i 4r^2}{2\left(d^2 + 4r^2\right)^{3/2}}$$

Magnetic field at O due to smaller coil X is

$$B_X = \frac{\mu_0 i r^2}{2\left[r^2 + \left(\frac{d}{2}\right)^2\right]^{3/2}}$$

$$B_X = \frac{\mu_0 i 8r^2}{2\left(d^2 + 4r^2\right)^{3/2}}$$

$$= \frac{B_Y}{B_X} = \frac{1}{2}$$

Final Answer : (3)

11) **Explaining :** To Match the column as given

Concept : This question is based on

Di	Para	ferro
< 0	> 1	>> 1

□ & 1 + □ = μ_r

Solution :

A → s, B → q, C → p, D → r

Final Answer : (1)

12) **Explaining :** A hollow cylinderical wire carries a current I, having inner and outer radii 'R' and 2R respectively. Magnetic field at a point which 3R/2 distance away from its axis is :-

Concept : This question is based on $B_{mid} = \frac{\mu_0 I}{2\pi r}\left(\frac{r^2 - a^2}{b^2 - a^2}\right)$.

Solution :

$$B_{mid} = \frac{\mu_0 I}{2\pi\left(\frac{3R}{2}\right)}\left(\frac{\left(\frac{3R}{2}\right)^2 - R^2}{(2R)^2 - R^2}\right)$$

$$B_{mid} = \frac{5\mu_0 I}{36\pi R}$$

Final Answer : (3)

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13) **Explanation the Question:**
Proton has velocity with a component parallel to the field (along Y) and a perpendicular component; perpendicular part makes uniform circular motion, so overall path is a helix.

Concept: This question is based on Radius form $r = \frac{mv_\perp}{qB}$. Period from $T = \frac{2\pi m}{qB}$. Parallel velocity gives axial advance.
Formula:
$v_x = v \cos 60 = 1.0 \times 10^6$ m/s (perpendicular to B)
$v_{||} = v \sin 60 = 1.732 \times 10^6$ m/s (parallel to B)

$$r = \frac{mv_\perp}{qB}$$

$$T = \frac{2\pi m}{qB}$$

Solution:

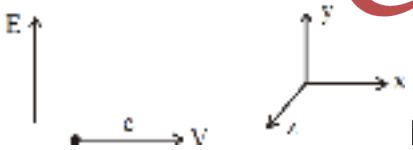
$$r = \frac{1.67 \times 10^{-27} \times 1.0 \times 10^6}{1.6 \times 10^{-19} \times 0.104} \approx 0.10\text{m}$$

$$T = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.104} \approx 2\pi \times 10^{-7}\text{s}.$$

Final Answer : Option (3)

14)

Question Explanation: We need a magnetic field so the electron continues straight along +x.
Concept: Velocity selector: qE=qvB
Solution :



$\vec{B}$ must be in +z axis.

$\vec{V} = 6 \times 10^6 \hat{i}$
$\vec{E} = 300\hat{j}$ V/cm $= 3 \times 10^4$ V/m
$q\vec{E} + q\vec{V} \times \vec{B} = 0$
E = VB

$$B = \frac{E}{V} = \frac{3 \times 10^4}{6 \times 10^6} = 5 \times 10^{-3}\text{T}$$

Final Answer : Option (1)

15) **Explain Question:**
A semi-circular wire of radius R carrying current I is placed in a magnetic field perpendicular to the plane. On the left of line XX' magnetic field is B_0 on the right it is $2B_0$. The net magnetic force on the semi-circular wire will be.

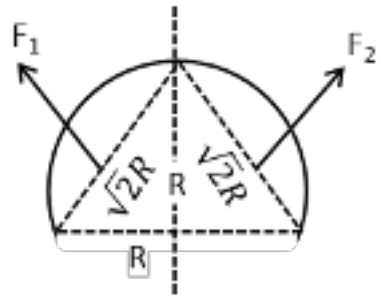
Concept: Magnetic force on a current-carrying conductor in a non-uniform magnetic field.

Formula:
For an arc in uniform magnetic field:
F = IB × (chord length vector)

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For a semicircle of radius R : Chord = $\sqrt{2}$R.

Solution:
The semicircle is divided into two equal quarter circles by line XX';



Left side (magnetic field B_0):
Force magnitude
$F_1 = I B_0 \times (\sqrt{2}R)$
Right side (magnetic field $2B_0$)
:$F_2 = I (2B_0) \times (\sqrt{2}R) = 2\sqrt{2}\, I B_0 R$
Thus the two forces are perpendicular.
Resultant force:

$$F = \sqrt{(F_1^2 + F_2^2)}$$

$$F = \sqrt{(\sqrt{2}B_0IR)^2 + (2\sqrt{2}B_0IR)^2}$$

$$F = \sqrt{10}B_0IR$$

Final answer: Option (3)

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16) **Explaining :** The magnetic field at a point X on the axis of a small bar magnet, is equal to the field at a point Y on the equator of the same magnet. The ratio of the distances of point X and Y from the centre of the magnet is :-
Concept : This question is based on magnetic field due to bar magnet.
Solution : $B_x = B_y$

$$\Rightarrow \frac{\mu_0}{4\pi}\frac{2M}{x^3} = \frac{\mu_0}{4\pi}\frac{M}{y^3}$$

$$\Rightarrow \left(\frac{x}{y}\right)^3 = 2 \Rightarrow \frac{x}{y} = (2)^{1/3}$$

Final Answer : (4)

17) **Explain Question:** Asking about the magnetic moment of a circular loop for give magnetic field at it's center.
Concept: Magnetic field of a loop.
Solution:
For a circular current carrying loop, magnetic filed at it's center & magnetic moment is given as.

$$B = \frac{\mu_0 i}{2R} \quad \text{--------(1)}$$

M = iA --------(2)
 A = πR^2

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$$R = \sqrt{\frac{A}{\pi}}$$

Eq. (2)/Eq. (1)

$$\frac{M}{B} = \frac{A\,2R}{\mu_0}$$

$$M = \frac{2BA}{\mu_0}\sqrt{\frac{A}{\pi}}$$

Final answer: Option (4)

18) **Explain Question:**
question asking about Torque (T) on a magnetic diploe in a magnetic field.

Concept:
Torque on a magnetic dipole. in a magnetic field.

Solution:
Formula: Using the formula for torque- $\vec{\tau} = \vec{M} \times \vec{B}$

$\vec{M} = -25\hat{j}A - m^2$, $\vec{B} = 5\hat{i} + 3\hat{j}T$

Now, let's calculate the cross products:

$$\vec{\tau} = \left(-25\hat{j}\right) \times \left(5\hat{i} + 3\hat{j}\right)$$

$$\vec{\tau} = -25\left[5\left(\hat{j} \times \hat{i}\right) + 3\left(\hat{j} \times \hat{j}\right)\right]$$

$\hat{j} \times \hat{i} = -\hat{k}$, $\hat{j} \times \hat{j} = 0$

$$\vec{\tau} = -25\left[5\left(-\hat{k}\right)\right]$$

$$\vec{\tau} = 125\hat{k}\ Nm$$

Final Answer: option(2)

19) **Explaining :** Match column-I with corresponding results in column-II.
Concept : This question is based on magnetic dipole in external uniform magnetic field.
Solution : Torque on a dipole $\tau = \mu B\sin\theta$
Potential energy of dipole $U = -\mu B\cos\theta$
If μ and $\vec{B}$ are parallel $\theta = 0°$; $\sin 0° = 0$, $\tau = 0$; $U_{min} = -\mu B$
If μ and are perpendicular $\theta = 90°$; $\sin 90° = 1$ $\tau_{max} = \mu B$; $U = 0$
If $\theta = 180°$ $\tau = 0$; $U_{max} = \mu B$
Final Answer : (4)

20)

QUESTION EXPLANATION
A solenoid has a core material of relative permeability $\mu_r = 400$. It carries a current of 2 A and has 1000 turns per metre. We have to calculate the magnetising field H and the intensity of magnetization I.

CONCEPT
Magnetisation of a solenoid with magnetic core

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SOLUTION
Magnetising field,
$H = n \times i$
$H = 1000 \times 2 = 2 \times 10^3 \text{ A m}^{-1}$

Relative permeability,
$\mu_r = 1 + \chi$

So,
$\chi = \mu_r - 1 = 400 - 1 = 399$

Intensity of magnetization,
$I = \chi \times H$
$I = 399 \times 2 \times 10^3$
$I = 798 \times 10^3 \text{ A m}^{-1}$

FINAL ANSWER: Option (1)

21) **Explaining :** half deflection method,
Concept : This question is based on half deflection method.
Solution : In half deflection method, a high resistance box is connected in series with the battery so that the deflection of the galvanometer is brought within the scale.
Final Answer : (1)

22) **Explaining :** To determine the ratio of the ratios of current sensitivity to voltage sensitivity of M_2 and M_1.
Concept : This question is based on sensitivity of moving coil galvanometer.
Solution : $\frac{\text{Current sensitivity}}{\text{Voltage sensitivity}} =$ Resistance of moving coil galvanometer

$$\frac{\left(\frac{S_I}{S_V}\right)_2}{\left(\frac{S_I}{S_V}\right)_1} = \frac{R_2}{R_1} = \frac{14}{10} = 1.4$$

Final Answer : (1)

23) **Explaining :** To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ. If a shunt of resistance S is needed to get half deflection then G, R and S are related by the equation :
Concept : This question is based on half-deflection method.
Solution :

$$i_1 = \frac{E}{R+G}$$

$$i_2 = \frac{E}{R + \frac{GS}{G+S}} = \frac{E(G+S)}{RG+RS+GS}$$

$$\frac{i_1}{2}G = \left(i_2 + \frac{i_1}{2}\right)S$$

$$\frac{i_1}{2}(G+S) = \frac{E(G+S)}{RG+RS+GS}.S$$

$$\frac{1}{2(R+G)} = \frac{S}{RG+RS+GS}$$

$$2RS + 2SG = RG + RS + GS$$

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RS + GS = RG
S (R+G) = RG
Final Answer : (2)

24) **Explain Question :** A force is applied on a body of mass 2 kg which varies with displacement of body according to given F-s graph. If initial speed is 2 m/s and we need to find the final speed of the body.
Concept : This question is based on work-energy theorem, force-displacement graph.
Solution : Let apply W.E.T
$W = KE_f - KE_i$

$$\frac{1}{2} \times 25 \times 50 = \frac{1}{2} \times 5v_f^2 - \frac{1}{2} \times 5 \times (10)^2$$

$$625 = 5v_f^2 - 500$$

$$v_f^2 = 225$$

$$v_f = 15$$

object is finally move at 15 m/s
Final Answer: Option (1)

25)

Question Explanation: The question asks to find the work done to pull a hanging chain onto a table.
Concept: Work-Energy Principle / Potential Energy
Solution:
Total mass M = 6 kg, Total length L = 3m.
Hanging length h = 1 m.

Mass of hanging part $m = \frac{M}{L} \times h = \frac{6}{3} \times 1 = 2kg$.

|Loss in PE| = |Gain in KE|

$$|PE_i - PE_f| = |KE_f - KE_i|$$

$$|2 \times 10 \times 0.5 - 6 \times 10 \times 3.5| = \left|\frac{1}{2} \times 6 \times v^2 - 0\right|$$

$$v \approx 8m/s$$

Final Answer: Option (1)

26) **Explain the Question :**
We are given a graph of variation of potential v/s position. If its total energy is 30 J and particle is at point 'C' initially.
Concept : This question is based on energy of the particle.
Solution :
By the mechanical energy conservation.
Total ME = PE + KE
At point 'C' 30 = – 30 + KE

$$\boxed{KE = 60J}$$

(1) For reaching at point E it should be cross the point 'D' which has 65 J more potential energy than 'C' which is greater than KE at point 'C' i.e. never reach at 'D'.
Hence option (3) may be possible
Final Answer : 3

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27) **Explaining :** To find type of equilibrium at position C.
Concept : This question is based on concept of potential energy
Solution :
$\frac{dF}{dx} < 0$ so stable equilibrium.
Final Answer : (1)

28) **Explaining :** To find work done by given force.
Concept : This question is based on concept of work done by variable force.
Solution :
$F = 8t$; $m.\frac{dv}{dt} = 8t$; $m\int_0^v dv = 8\int_0^t t\,dt$, $mv = 4t^2$
$mv = 4t^2$
$v = \frac{4t^2}{2} = 2t^2$
$v_i = 0 m/s$
$v_f = 8 m/s$
$WD = \Delta KE = \frac{1}{2}m(v_f^2 - v_i^2)$
$WD = \Delta KE = \frac{1}{2} \times 2\left((8)^2 - (0)^2\right)$
$= 64\ J$
Final Answer : (1)

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29) **Explaining :** To find distance travelled by particle before it stops
Concept : This question is based on concept of friction
Solution :
Applying work energy theorum
$W_g + W_N + W_f = K_f - K_i$
$mg(2r) + 0 - \mu mgx = 0 - \frac{1}{2}m(2gr)$
$mgx = 3\ mgr$
$x = \frac{3r}{\mu} = 6m$
Final Answer : (3)

30) **Explaining :** Power is the rate of doing work. When lifting water, the work done is against gravity (Potential Energy).
Concept: Work (W) = mgh
Output Power (P_{out}) = Work / Time
Input Power (P_{in}) = Output Power / Efficiency
Solution :
$P_{out} = mgh / t$
$P_{out} = (600 \times 10 \times 20) / 60$
$P_{out} = 2000\ W$

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Efficiency = P_{out} / P_{in}
P_{in} = P_{out} / Efficiency
P_{in} = 2000 / 0.80
P_{in} = 2500 W
Final Answer : (2)

31) **Explaining :** To find change in KE
Concept : This question is based on concept of power
Solution :

$$W_{net} = \int Pdt$$

Also $w_{net} = \Delta KE$

$$\Delta KE = \int Pdt$$

$$= \int_2^4 (3t^2 - 2t + 1)dt = \left[3\frac{t^3}{3} - 2\frac{t^2}{2} + t\right]_2^4$$

$$= \left[t^3 - t^2 + t\right]_2^4 = [4^3 - 4^2 + 4] - [2^3 - 2^2 + 2]$$

ΔKE = 46J
Final Answer : (1)

32) **Explaining :** To find spring constant K.
Concept : This question is based on concept of maximum tension in cord.
Solution :
Note that in this problem, L = H.

$$mgH = \frac{1}{2}kh^2$$

$$4mg = kh$$

$$\frac{mgH}{4mg} = \frac{kh.h}{2kh} \Rightarrow h = \frac{H}{2}$$

$$k = \frac{4mg}{h} = \frac{8mg}{H}$$

Final Answer : (3)

33) **Explaining :** To find average power to be delivered.
Concept : This question is based on concept of average power
Solution :
u = 10; $\quad$ v = 20

$$W = \frac{1}{2} \times 500[20^2 - 10^2]$$

$$\text{Power} = \frac{500 \times 30 \times 10}{2 \times 60} = 1250 \text{ W}$$

Final Answer : (4)

34) **Explaining :** To find the maximum speed (in m/s).
Concept : This question is based on concept of potential energy.
Solution :

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Potential energy $U = \frac{x^4}{4} - \frac{x^2}{2}$

For maximum kinetic energy, potential energy of a particle should be minimum.

For minimum value of U, $\frac{dU}{dx} = 0$ and $\frac{d^2U}{dx^2} > 0$

After calculation at $x = \pm 1$

$U_{min} = -\frac{1}{4}J$

$TE = KE_{max} + PE_{min}$

$\Rightarrow KE_{max} = TE - PE_{min}$

$\Rightarrow \frac{1}{2}mv_{max}^2 = 2 - \left(-\frac{1}{4}\right)$

$v_{max} = \frac{3}{\sqrt{2}}m/s$

Final Answer : (4)

35) **Explaining :** To find work done by normal.

Concept : This question is based on concept of work done by constant force.

Solution :

Normal Reaction (N)

mg – N = ma

N = m(g – a)

N = 40 (10 – 10/4)

N = 300 Newton

W = N d cos(180)

W = 300×10 ×(–1)

W = -3000 Joules

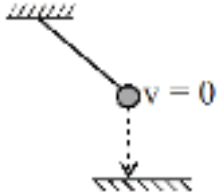
Final Answer : (3)

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36) **Explaining :** To find trajectory possible for the bob.

Concept : This question is based on concept of vertical circular motion.

Solution :



Final Answer : (3)

37) **Explaining :** To find ratio of tension.

Concept : This question is based on concept of tension in string

Solution :

$T_2 = m_{2\times}\omega^2 \times r_2$

$T_2 = 4 \times 1.5 \times \omega^2$

$T_2 = 6\ \omega^2$

$T_1 = m_{1\times}\omega^2 \times r_1 + T_2$

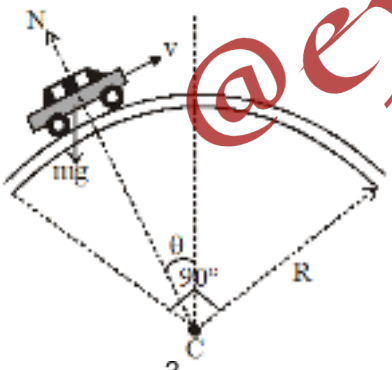
$T_1 = 2 \times 0.5 \times \omega^2 + T_2$

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$T_1 = 7\ \omega^2$
Ratio = T_1 / T_2
Ratio = $7\ \omega^2 / 6\ \omega^2$
Ratio = 7 / 6
Final Answer : (2)

38) **Explaining :** To find speed & sec.
Concept : This question is based on concept of horizontal cm.
Solution :
Angular speed of table = $\omega = \frac{v}{r} = \frac{20}{r} = \frac{v}{r/2}$
v = 10 cm/s
$r\omega^2 = 20$
$\omega^2 = \frac{20}{r} = \frac{a_c}{r/2}$
$a_c = 10 cm/s^2$
Final Answer : (1)

39) **Explaining :** To find minimum possible speed
Concept : This question is based on concept of contact force.
Solution :
Let the car losses the contact at angle θ with the vertical
$mg\cos\theta - N = \frac{mv^2}{R}$



$N = mg\cos\theta - \frac{mv^2}{R}$ .....(1)
for losing the contact N =0
$\Rightarrow v = \sqrt{Rg\cos\theta}$ from (1)
for minimum speed, cosθ should be minimum so that θ should be maximum.
$\theta_{max} = 45° \Rightarrow \cos 45° = \frac{1}{\sqrt{2}}$
$v_{min} = \left(\frac{Rg}{\sqrt{2}}\right)^{\frac{1}{2}}$ Ans.
So that if car cannot lose the contact at initial or final point, car cannot lose the contact anywhere.
Final Answer : (2)

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40) **Explaining :** To find no. of revolution.
Concept : This question is based on concept of angular velocity.
Solution :
$\omega = \omega_0 + \alpha t$
$20 = 0 + \alpha(5)$
$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$
$(20)^2 = 0 + 2(4)\ \Delta\theta$
$\Delta\theta = \frac{400}{8} = 50$ rad
$n = \frac{\Delta\theta}{2\pi} = \frac{25}{\pi}$rev.
Final Answer : (3)

41) **Question Explanation :** A cyclist rides on a circular road, find the banking angle to prevent skidding at given speed and radius.

Concept : This question is based on banking.

Solution :
$\tan\theta = \frac{v^2}{rg}$
$\tan\theta = \frac{14\sqrt{3} \times 14\sqrt{3}}{20\sqrt{3} \times 9.8} = \sqrt{3}$
$\therefore \theta = 60°$

Final Answer : option (4)

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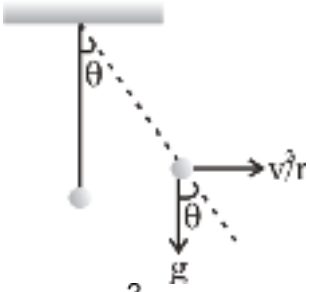
42) **Explaining :** To find acceleration of 'p'.
Concept : This question is based on concept of acceleration of 'cm'.
Solution :
As $S = t^3 + 6$
$\frac{ds}{dt} = 3t^2 = v$
$a_t = \frac{dv}{dt} = 6t$
at t = 2 sec
$|\vec{a}| = \sqrt{a_c^2 + a_t^2}$
$= \sqrt{\left(\frac{v^2}{R}\right)^2 + a_t^2} = \sqrt{\left(\frac{9t^4}{R}\right)^2 + \left(\frac{dv}{dt}\right)^2}$
$= \sqrt{(9)^2 + 144}$
$|\vec{a}| = \sqrt{225} = 15 m/s^2$
Final Answer : (4)

43) **Explaining :** To find normal reaction
Concept : This question is based on concept of vert. cm

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Solution :
$m\ g\ h = (1/2)\ m\ v^2$
$\mathbf{v^2} = 2 \times g \times h$
$\mathbf{v^2} = 2 \times 10 \times 5$
$\mathbf{v^2 = 100}$ (v = 10 m/s)
$\mathbf{N - mg = mv^2 / R}$
$N = mg + (m\ v^2 / R)$
$N = 20 + 100$
N = 120 Newton
Final Answer : (3)

44) **Explaining :** To find angle made by the rod
Concept : This question is based on concept of vertical circular motion
Solution :



$$\tan\theta = \frac{v^2}{rg}$$

$$\therefore \theta = \tan^{-1}\frac{v^2}{rg}$$

$$= \tan^{-1}\left(\frac{10 \times 10}{10 \times 10}\right)$$

$\therefore \theta = \tan^{-1}(1) = 45°$
Final Answer : (3)

45) **Explaining :** To find centripetal acceleration
Concept : This question is based on concept of Horizontal circular motion.
Solution :
Tension T in the string will provide centripetal force

$$\Rightarrow \frac{mv^2}{\ell} = T$$

Also, tension T is provided by the hanging ball of mass m,

$\Rightarrow T = mg$

$$mg = \frac{mv^2}{\ell} \Rightarrow \frac{v^2}{\ell} = g$$

It is the centripetal acceleration of a moving ball
Final Answer : (2)

CHEMISTRY

46)

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Explaining : The oxidation potential of hydrogen electrode at pH = 10, 1 atm and 25°C is :-
Concept : This question is based on the Nernst equation $E = E° - (RT/nF) \times \ln(Q)$
Solution :
$H_2 \longrightarrow 2H^+ + 2e^-$
For the hydrogen electrod. Since P_{H_2} = 1 atm, $[H^+] = 10^{(-pH)}$

$$E = \frac{0-0.0591}{2}\log[H^+]^2$$

$= -0.0591 \log 10^{-10} = 0.591$ V
Final Answer : (1)

47) **Explaining :** The standard reduction potential values of three metallic cations, X,Y,Z are 0.52, –3.03 and –1.18 V respectively. The order of reducing power of the corresponding metals is :-
Concept : This question is based on application of electrochemical series.
Solution : More negative or lower is the reduction potential, more is the reducing power.
Final Answer : (1)

48)

Explaining : Identify the correct statements.
Concept : This question is based on qualitative analysis of electrolysis.
Solution :
Statement-I : For electrolysis of aq. NaCl Product obtained are
Anode :- $2Cl^- \rightarrow Cl_{2(g)} + 2e^-$
Cathode :- $2H_2O_{(l)} + 2e^- \rightarrow H_{2(g)} + 2OH^-_{(aq.)}$
Statement-II : $E^o_{Cu^{+2}/Cu} > E^o_{Fe^{+2}/Fe}$
Iron is more reactive than copper.
Final Answer : (3)

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49) **Explaining :** The relationship between standard potential of cell and equilibrium constant is given by :-
Concept : This question is based on Nernst equation.
Solution :
Relevant Formulas: At equilibrium, when E = 0 and Q = K, the Nernst equation gives:

$$0 = E^\circ - \frac{0.0591}{n}\log K$$

Rearranging:

$$E^\circ = \frac{0.0591}{n}\log K$$

Final Answer : (2)

50)

Question Explanation
The question asks to determine the molar conductance of sodium nitrate $(NaNO_3)$ at infinite dilution, given the molar conductances of three other salts: silver nitrate $(AgNO_3)$, silver chloride $(AgCl)$, and sodium chloride (NaCl).
Concept

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Kohlrausch's Law

Solution

$\Lambda_m^\circ (NaNO_3) = \Lambda_m^\circ (AgNO_3) + \Lambda_m^\circ (NaCl) - \Lambda_m^\circ (AgCl)$

$\Lambda_m^\circ (NaNO_3) = 116.5 + 110.3 - 121.6$

$\Lambda_m^\circ (NaNO_3) = 226.8 - 121.6 = 105.2 \Omega^{-1}\ cm^2\ mol^{-1}$

Final Answer: (2)

51) **Explaining :** The conductivity of 0.01 M solution is $10^{-4}\ \Omega^{-1}\ cm^{-1}$. Find its molar conductance (in terms of $\Omega^{-1}\ cm^2\ mol^{-1}$) :-

Concept : This question is based on molar conductance.

Solution :

$$\Lambda_m = \frac{\kappa \times 1000}{M}$$

$$= \frac{10^{-4} \times 1000}{10^{-2}} = 10$$ **Final Answer :** (1)

52) **Explaining :** Which of the following representation is correct?

Concept : This question is based on cell representation.

Solution : $Mg + 2Ag^+ \rightarrow Mg^{+2} + 2Ag$; $Q = \frac{[Mg^{+2}]}{[Ag^+]^2}$

Final Answer : (3)

53) **Explaining :** The specific conductance of a 0.1 N KCl solution at 23°C is 0.012 $ohm^{-1}\ cm^{-1}$. The resistance of cell containing the solution at the same temperature was found to be 55 ohm. The cell constant will be

Concept : This question is based on cell constant.

Solution :

$k = G \times G^*$

$$G^* = \frac{k}{G} = 0.012 \times 55 = 0.66$$

Final Answer : (2)

54)

Explaining : The ionization constant of a weak electrolyte is 25×10^{-6} while the equivalent conductance of its 0.01 M solution is 19.6 S $cm^2\ eq^{-1}$. The equivalent conductance of the electrolyte at infinite dilution (in S $cm^2\ eq^{-1}$) will be

Concept : This question is based on Kohlrausch law.

Solution :

$K = 25 \times 10^{-6}$ $\Lambda_{eq} = 19.6 S\ cm^2\ eq^{-1}$

$M = 0.01\ M$

$$K = \frac{c\alpha^2}{1-\alpha} \quad 1 - \alpha \simeq 1$$

$$\alpha = \sqrt{\frac{k}{C}} = \sqrt{\frac{25 \times 10^{-6}}{0.01}} = 5 \times 10^{-2}$$

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$$\alpha = \frac{\Lambda_{eq}}{\Lambda_{eq}^{\infty}} \quad 5 \times 10^{-2} = \frac{\Lambda_{eq}}{\Lambda_{eq}^{\infty}}$$

$$\Lambda_{eq}^{\infty} = \frac{19.6}{5 \times 10^{-2}} = 392$$

Final Answer : (3)

55) **Explaining :** E° for the cell,
$Zn \mid Zn^{2+}(aq) \parallel Cu^{+2}(aq) \mid Cu$ is 1.10 V at 25°C. The equilibrium constant for the cell reaction

$$Zn + Cu^{2+}(aq) \rightleftharpoons Cu + Zn^{2+}(aq)$$

is of the order of
Concept : This question is based on Nernst equation.
Solution :

$$E^{o} = \frac{0.059}{n} \log K_c \; ; \; 1.1 = \frac{0.059}{2} \log K_c$$

$$K_c \approx 10^{37.28}$$

Final Answer : (2)

56) **Explaining :** $2Ce^{4+} + Co \rightarrow 2Ce^{3+} + Co^{2+}$; $E^{o}_{Cell} = 1.89V$, $E^{o}_{Co/Co^{+2}} = 0.277V$.
Hence, $E^{o}_{Ce^{4+}/Ce^{3+}}$ is :
Concept : This question is based on E_{cell} calculation.
Solution :

$$E^{o}_{cell} = (SRP)_{cathode} - (SRP)_{anode}$$

$$1.89 = (SRP)_c - (-0.277)$$

$(SRP)_c = 1.613$ V
Final Answer : (2)

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57) **Explaining :** Match the column.
Concept : This question is based on Faraday's law.
Solution :
(A) $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$
1 mol H_2O will req. 2F charge
(B) $8H^+ + MnO_4^- + 5e^- \rightarrow Mn^{+2} + 4H_2O$
1 mole MnO_4^- will req. 5F charge.
(C) $CaCl_2 + 2e^- \rightarrow Ca + 2Cl^{-1}$
1.5 mole Ca will req. 3F charge.
(D) $2FeO + H_2O \rightarrow Fe_2O_3 + 2H^+ + 2e^-$
1 mole FeO will req. 1 F charge.
(A)-(II), (B)-(IV), (C)-(I), (D)-(III)
Final Answer : (1)

58) **Explaining :** In a galvanic cell.
Concept : This question is based on galvanic cell.
Solution : Fact based.
Final Answer : (1)

59) **Explaining :** Function of salt bridge.

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Concept : This question is based on function of salt bridge.
Solution :
It completes the electrical circuit with electrons flowing from one electrode to the other through external wires and a flow of ions between the two compartments through salt-bridge. It minimizes the liquid-liquid junction potential
Final Answer : (3)

60) **Explaining :** Function of salt bridge.
Concept : This question is based on function of salt bridge.
Solution : Fact based.
Final Answer : (3)

61)

Explaining : E.M.F. of the cell
$2Ag^{+} + Cu \rightarrow Cu^{+2} + 2Ag$
[Given: $E^{\circ}_{Ag|Ag^{+}}$ = –0.8 V ; $E^{\circ}_{Cu^{+2}|Cu}$ = 0.3 V]
Concept : This question is based on E^{o}_{cell} calculation.
Solution : $E^{\circ}_{cell} = E^{\circ}_{Ag^{+}|Ag} - E^{\circ}_{Cu^{2+}|Cu} = +0.8V - 0.3V = 0.5V$
Final Answer : (2)

62) **Explaining :** Given : E° (Cu^{2+}/Cu) = 0.337 V and E° (Sn^{2+} /Sn) = –0.136 V. Which of the following statements is correct ?
Concept : This question is based on application of ECS.
Solution : Lesser the SRP of metal higher the reactivity.
Final Answer : (1)

63) **Explaining :** Consider the cell potential $E^{o}_{Mg^{2+}|Mg}$ = –2.37 V and $E^{o}_{Fe^{3+}|Fe}$ = –0.04 V. The best reducing agent would be :
Concept : This question is based on application of ECS.
Solution : R.A. itself gets oxidised easily, i.e., Mg.
Final Answer : (3)

64)

Explaining : The number of Faradays(F) required to produce 20 g of calcium from molten $CaCl_2$ (Atomic mass of Ca = 40 g mol^{-1}) is :
Concept : This question is based on Faraday's law.
Solution :
$Ca^{+2} + 2e^{-} \rightarrow Ca(s)$
v.f. = 2
As per faraday's 1st law
Charge passed in faraday = g.eq of product

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$= \frac{20}{40} \times 2 = 1F$

Final Answer : (2)

65) **Explaining :** When 0.1 mol MnO_4^{2-} is oxidised the quantity of electricity required to completely oxidise MnO_4^{2-} to MnO_4^- is :-

Concept : This question is based on Faraday's law.

Solution :

Balancing Charge

$MnO_4^{-2} \longrightarrow MnO_4^- + 1e^-$

$\downarrow$

$1mol \longrightarrow 1mol\ e^-$

$0.1mol\ MnO_4^{-2} \longrightarrow 0.1\ mol\ e^-$

$\Downarrow$

$0.1\ F$

$\Downarrow$

0.1×96500

$= 9650\ C$

Final Answer : (3)

66)

Explaining : Strong electrolytes are those which :

Concept : This question is based on strong electrolytes.

Solution : Strong electrolytes are those for which degree of dissociation (α) is 100% at any concentration.

Final Answer : (3)

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67) **Explaining :** The specific conductance of a 0.01M solution of KCl is 0.0014 $ohm^{-1}cm^{-1}$ at 25°C. Its equivalent conductance ($cm^2ohm^{-1}eq^{-1}$) is :

Concept : This question is based on equivalent conductivity.

Solution :

Given : specific conductance

$(\kappa) = 0.0014\,\Omega^{-1}\,cm^{-1}$

Conc. (M) = 0.01M

Now

$\lambda_{eq} = \frac{\kappa \times 1000}{N}$ n-factor for KCl = 1

So N = M = 0.01

$= \frac{0.0014 \times 1000}{0.01}$

$\boxed{\lambda_{eq} = 140}$

Final Answer : (1)

68) **Explaining :** The molar conductivities $\wedge^0_{NaOAc}$ and $\wedge^0_{HCl}$ at infinite dilution in water at 25°C are 91.0 and 426.2 S cm^2 mol^{-1} respectively. To calculate $\wedge^0_{HOAc}$ the additional value

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required is :
Concept : This question is based on Kohlrausch law.
Solution :
$\wedge^{o}_{HOAc} = \wedge^{o}_{H^+} + \wedge^{o}_{OAc^-}$
$= \wedge^{o}_{NaOAc} + \wedge^{o}_{HCl} - \wedge^{o}_{NaCl}$
To subtract molar conductance of Na^+ and Cl^- We required $\wedge^{0}_{NaCl}$. **Final Answer :** (1)

69) **Explaining :** The osmotic pressure of equimolar solution of $BaCl_2$, NaCl and glucose will be in the order :
Concept : This question is based on osmotic pressure.
Solution : If conc. is same then, $\pi \propto i$.
Final Answer : (2)

70) **Explaining :** Which one of the following aqueous solutions exhibit highest boiling point?
Concept : This question is based on elevation in boiling point.
Solution : $\Delta T_b \propto i \cdot m$
Final Answer : (1)

71) **Explaining :** Which of the following solutions can have boiling point less than that of both the individual components?
Concept : This question is based on Non-ideal solutions.
Solution : C_2H_5OH and H_2O form positive deviation solution and hence, minimum boiling azeotropic mixture.
Final Answer : (4)

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72)

Explaining : On mixing 10mL of acetone with 40mL of chloroform, the total volume of the solution is:
Concept : This question is based on Non-ideal solutions.
Solution : Negative deviation, $\Delta V_{mix} < 0$
Final Answer : (1)

73) **Explaining :** Statement type question.
Concept : This question is based on abnormal molecular mass.
Solution : Abnormal colligative properties. avg. molecular mass increases due to association
Final Answer : (1)

74) **Explaining :** Molarity of 720 g of pure water -
Concept : This question is based on molarity.
Solution : Molarity of pure water is always 55.5 M
Final Answer : (3)

75) **Explaining :** Osmotic pressure of 30% solution of glucose is 1.20 atm and that of 3.42% solution of cane-sugar is 2.5 atm. The osmotic pressure of the mixture containing equal

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volumes of the two solutions will be :
Concept : This question is based on osmotic pressure.
Solution : $\pi = \frac{\pi_1 V_1 + \pi_2 V_2}{V_1 + V_2} = 1.85$ atm
Final Answer : (3)

76) **Explaining :** Which of the following is independent of temperature :
Concept : This question is based on concentration terms.
Solution : $m = \frac{\text{No. of moles}}{\text{Mass of solvent (kg)}}$
Final Answer : (3)

77) **Explaining :** Molarity represents number of moles of solute present in :
Concept : This question is based on concentration terms.
Solution : $M = \frac{n}{V_{(\ell)}}$
Final Answer : (3)

78) **Explaining :** Which of the following is not a colligative property ?
Concept : This question is based on colligative property.
Solution : Fact.
Final Answer : (4)

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79)

Explaining : Correct statement for osmosis
Concept : This question is based on osmosis
Solution : 0.1 M glucose is isotonic to 0.1 M urea hence there will be no net movement across the membrane.
Final Answer : (1)

80)

Explaining : Evaluate given Assertion and Reason.
Concept : This question is based on osmotic pressure (i > 1)
Solution : 0.1 M solution of NaCl has greater osmotic pressure. than 0.1 M solution of glucose at same temperature because In solution, NaCl dissociates to produce more number of particles.
Final Answer : (1)

81)

Explaining : Calculate amount of ice separated on cooling a solution ?
Concept : This question is based on depression in freezing point.
Solution :

$9.3 = 1.86 \times \frac{\frac{50}{62}}{w} \times 1000 \Rightarrow w = 161.3$ g

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 Amount of ice = 200 – 161.3 = 38.7 g
Final Answer : (1)

82)

Explaining : Find the incorrect statement
Concept : This question is based on multiconcepts
Solution : Depression in freezing point, not only depends on the concentration of solution but also on the cryoscopic constant. Cryoscopic constant is different for different solvent.
Final Answer : (2)

83)

Explaining : Choose the correct option for the given curve.
Concept : This question is based on freezing point curve.
Solution :
Point (B) denotes freezing point of pure solvent and point (D) denotes freezing point of solution. Thus, the curve (B-D-E) denotes vapour pressure curve for frozen solvent.
Final Answer : (3)

84)

Explaining : Find Osmotic pressure of the resultant solution.
Concept : This question is based on osmotic pressure.
Solution :

$$\pi V = \left(\frac{\omega_1}{m_1} + \frac{\omega_2}{m_2}\right) RT$$

$$\pi \times 2 = \left(\frac{18}{180} + \frac{34.2}{342}\right) \times 0.082 \times 300$$

$$\pi \times 2 = 0.2 \times 0.082 \times 300$$

$$\pi = 2.46 \text{ atm}$$

Final Answer : (2)

85)

Explaining : Calculate the value of ebullioscopic constant of acetone.
Concept : This question is based on ebullioscopic constant.
Solution :

$$\Delta T = K_b \times \frac{w_B \times 1000}{m_B \times w_A}$$

$$0.3 = K_b \times \frac{7 \times 1000}{210 \times 350}$$

$$K_b = \frac{0.3 \times 210 \times 350}{7 \times 1000} = 3.15 \text{ K kg mol}^{-1}$$

Final Answer : (2)

86) **Explaining :** The heat of neutratization of any strong acid and a strong base is always constant. This is because :
Concept : This question is based on heat of neutralization.

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Solution : Both the acid and base undergo complete ionization
Final Answer : (1)

87) **Explaining :** The equivalent weight of sodium carbonate [Na_2CO_3] is
Concept : This question is based on equivalent weight.
Solution : $$\text{Eq. weight} = \frac{\text{Molecular weight}}{\text{Valency factor}} = \frac{106}{2} = 53\,\text{g/eq.}$$
Final Answer : (2)

88)

Explaining : The number of gms of solute per 100 ml of solution is known as
Concept : This question is based on concentration terms.
Solution : % weight by volume
Final Answer : (3)

89) **Explaining :** Match the column.
Concept : This question is based on titration.
Solution : 1-R, 2-S, 3-Q, 4-P
Final Answer : (3)

90) **Explaining :** Which of the following is not a lyophilic colloid?
Concept : This question is based on lyophilic colloid.
Solution : Fog is an aerosol in which liquid is dispersed in gas and not a lyophilic colloid.
Final Answer : (3)

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91) **Solution :** Histones are enriched Lysine & Arginine.
Final Answer : (1)

92) **Solution :** George Gamow proposed that the genetic code for amino acids should be made up of three nucleotides.
Final Answer : (1)

93) **Solution :** ρ (rho) factor is important for termination of transcription.
Final Answer : (3)

94) **Solution :** 23 × 3.4 Å = 78.2 Å
Final Answer : (3)

95) **Solution :** S^{35} is found in protein coat of virus.
Final Answer : (1)

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96) **Solution :** Between N-Base and pentose sugar.
Final Answer : (1)

97) **Solution :** DNase inhibits transformation.
Final Answer : (4)

98) **Solution :** Pitch or helix length is 3.4 nm or 34Å.
Final Answer : (1)

99) **Solution :** Base pairing between two strands.
Final Answer : (1)

100) **Solution :** One purine always comes opposite to a pyrimidine
Final Answer : (3)

101) **Solution :** Genetic material should not change with any of these conditions.
Final Answer : (4)

102) **Solution :** 4.6×10^6 bp nitrogen base pairs are there in the DNA of *E.coli.*
Final Answer : (3)

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103) **Solution :** A cytidine is formed through N-glycosidic linkage by joining ribose sugar with cytosine.
Final Answer : (3)

104) **Solution :** 1000 nucleotides = 500 bp.
So, $2n - 2 = 2 \times 500 - 2 = 1000 - 2 = 998$
Final Answer : (3)

105) **Solution :** T = 15%, then A = 15%
A + G = 50%. So, 15% + G = 50% ,
G = 50 – 15 = 35%
Final Answer : (3)

106) **Solution :** A = Phosphate, B = Deoxyribose, C = Purine, D = Nucleoside
Final Answer : (1)

107) **Solution :** Nucleosome structure observed as bead on string during packaging of DNA.
Final Answer : (3)

108) **Solution :** RNA primer is synthesised by primase (RNA polymerase like enzyme).
Final Answer : (2)

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109) **Solution :** I-c, II-d, III-a, IV-b
Taylor — *Vicia faba*
Hershey and Chase — Bacteriophage
N^{15} was used by Meselson and Stahl
Griffth — *Streptococcus pneumoniae*
Final Answer : (3)

110) **Solution :** Out of 16, 2 DNA molecules are hybrid and 14 are heavy.
Final Answer : (3)

111) **Solution :** DNA synthesis always takes place in the direction of 5' → 3'.
Final Answer : (2)

112) **Solution :** Nucleic acid in the sample is single stranded DNA.
Final Answer : (1)

113) **Solution :** Degenerate, Non-overlapping, Non- ambiguous
Final Answer : (4)

114) **Solution :** Lagging strand has many okazaki fragments.
Final Answer : (2)

115) **Solution :** This indicates RNA world.
Final Answer : (3)

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116) **Solution :** 'i' stands for inhibitor, not inducer.
Final Answer : (4)

117) **Solution :** VNTRs show high degree of polymorphism among individuals.
Final Answer : (3)

118) **Explanation:**
Nucleotides: Each base pair consists of two nucleotides (one on each strand). Therefore, 4000 base pairs contain 4000 × 2 = 8000 nucleotides.
Sugar molecules: Each nucleotide contains one sugar molecule. Thus, there will be **8000** sugar molecules.
Phosphate molecules: Each nucleotide also contains one phosphate molecule. Therefore, there will be 8000 phosphate molecules.
Final Answer : (4)

119) **Solution :** Coding strand is same as RNA except T in place of U.
Final Answer : (1)

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120) **Solution :** Frame shift mutation will occur.
Final Answer : (3)

121) **Solution :** b = blending; c = centrifugation; a = infection
Final Answer : (3)

122) **Solution :** Living R-type bacteria changed into S-type.
Final Answer : (1)

123) **Solution :** Chemically, RNA is **more** reactive and **less** stable as compared to DNA.
Final Answer : (3)

124) **Solution :** RNA mutates as faster rate than DNA.
Final Answer : (3)

125) **Solution :** Non-histone chromosomal protein is required for higher level of DNA packaging.
Final Answer : (3)

126) **Solution :** $\frac{6.8 \times 10^{-3}\ \text{m}}{0.34 \times 10^{-9}\ \text{m}} = 2 \times 10^{7}\ \text{bp}$
Final Answer : (2)

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127) **Solution :** Euchromatin is active and heterochromatin is inactive.
Final Answer : (2)

128) **Solution :** A is adenine and found in both RNA and DNA.
Final Answer : (1)

129) **Solution :** A = repressor mRNA, B = inducer, C = permease, D = transacetylase, E = operator
Final Answer : (2)

130) **Solution :** It is prokaryotic transcription and involves sigma and rho factors.
Final Answer : (1)

131) **Question Explanation**: The question asks for the identification of A, B, and C in the prokaryotic transcription diagram.
Concept: Transcription in Prokaryotes
Solution:
RNA polymerase (A) binds to DNA with the help of sigma factor (B) at the promoter.
Sigma factor assists RNA polymerase in recognizing the promoter.
Rho factor (C) binds RNA to terminate transcription by releasing the RNA from RNA

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polymerase.
Final Answer: **(4)**

132) **Question Explanation:**Anticodon pairs with mRNA codon.
Concept: Codon-Anticodon Pairing
Solution:
mRNA: 5′ AUG GUG CUC AAA 3′
Codons → AUG, GUG, CUC, AAA
Matching anticodons (complementary and antiparallel):
AUG → **UAC** (c)
GUG → **CAC** (d)
CUC → **GAG** (b)
AAA → **UUU** (a) Correct order: c, d, b, a
Final Answer:3
c, d, b, a

133)

Question Explanation: The question asks whether the reason given correctly explains why the genetic code is nearly universal.

Concept: Genetic Code

Solution:
Genetic code is nearly universal because it is the same for almost all organisms, with exceptions in mitochondria and some protozoans.
Both the assertion and reason are correct, and the reason correctly explains the assertion.

Final Answer: Option (4)

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134) **Solution :** Mutations are alterations in DNA sequences.
Final Answer : (1)

135) **Question Explanation:** Evaluation of Assertion and Reason regarding the location and function of UTRs.
Concept: Structure and Function of mRNA UTRs
Solution: Assertion: True. UTRs (Untranslated Regions) are indeed located at the 5'-end (before the start codon) and the 3'-end (after the stop codon) of the mRNA molecule. They are not translated into protein.Reason: False. UTRs are sequences on the mRNA that are crucial for the efficient translation (protein synthesis) process, by affecting mRNA stability and regulating ribosome binding. They are not primarily required for transcription (DNA to mRNA synthesis), which involves promoter and terminator sequences on the DNA.Since the Assertion is True and the Reason is False, Option 3 is the correct choice.
Final Answer: (3)

136) **Explanation:** Mendel studied seven pairs of contrasting traits, which included seed shape, seed colour, flower colour, pod shape, pod colour, flower position, and stem height. Leaf size was not studied.

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Final Answer : (3)

137) **Explanation:** In the F_2 generation, the proportion of plants that were dwarf was $1/4^{th}$, while $3/4^{th}$ of the plants were tall, giving a phenotypic ratio of 3 : 1.
Final Answer : (2)

138) **Explanation:** In Snapdragon, the F_2 genotypic ratio is 1 (RR) : 2 (Rr) : 1 (rr), which is exactly the same as the expected Mendelian monohybrid genotypic ratio.
Final Answer : (3)

139) **Explanation:** Grasshopper is an example of XO type of sex determination in which the males have only one X-chromosome besides the autosomes, whereas females have a pair of X-chromosomes.
Final Answer : (2)

140) **Explanation:** The defect is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule.
Final Answer : (3)

141) **Explanation:** Mendel crossed the tall plant from F_2 with a dwarf plant to determine its genotype, calling it a test cross. The progenies of such a cross can easily be analysed to predict the genotype of the test organism.
Final Answer : (1)

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142) **Explanation:** Co-dominance is when the F_1 generation resembles both parents. In ABO blood groups, when I^A and I^B are present together, they both express their own types of sugars due to co-dominance.
Final Answer : (1)

143) **Explanation:** Genes on the same chromosome do not always assort independently; they show linkage. Morgan observed that linked genes have a higher proportion of parental combinations.
Final Answer : (4)

144) **Explanation:** A pleiotropic gene exhibits multiple phenotypic expressions. Phenylketonuria is caused by a single gene mutation but results in multiple phenotypic symptoms like mental retardation and reduced pigmentation.
Final Answer : (1)

145) **Explanation:** Since females produce only X-bearing ova, the fertilization by either an X-bearing sperm (resulting in a female) or a Y-bearing sperm (resulting in a male) determines the sex of the offspring.
Final Answer : (1)

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146) **Explanation:** Thalassemia differs from sickle-cell anaemia in that the former is a quantitative problem of synthesising too few globin molecules while the latter is a qualitative problem of synthesising an incorrectly functioning globin.
Final Answer : (1)

147) **Explanation:** Down's syndrome is trisomy of chromosome 21. However, Turner's syndrome results due to the loss of an X chromosome in human females (45, XO), not a gain.
Final Answer : (3)

148) **Explanation:** In birds, the total number of chromosomes is the same in males and females. However, birds exhibit *female* heterogamety (they produce two different types of gametes, Z and W), not male heterogamety.
Final Answer : (3)

149) **Explanation:** Mendel published his work in 1865, and it remained unrecognised until 1900 when three scientists (de Vries, Correns, and von Tschermak) independently rediscovered it.
Final Answer : (1)

150) **Explanation:** Polygenic traits are controlled by three or more genes. Human skin colour is a classic example where the effect of each allele is additive to determine the phenotype.
Final Answer : (1)

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151) **Explanation:** Down's is trisomy 21. Klinefelter's has 47, XXY. Turner's is 45, XO. Sickle-cell is due to Valine substitution for Glutamic acid.
Final Answer : (1)

152) **Explanation:** The recessive traits are: Seed shape - Wrinkled; Flower position - Terminal; Seed colour - Green; Pod shape - Constricted.
Final Answer : (1)

153) **Explanation:** Morgan coined linkage. Sutton & Boveri explained the chromosomal theory. Henking discovered the X body. Sturtevant developed genetic maps based on recombination frequencies.
Final Answer : (1)

154) **Explanation:** I^Ai produces blood type A. I^AI^B produces blood type AB. I^Bi produces blood type B. ii produces blood type O.
Final Answer : (1)

155) **Explanation:** Homozygous means identical alleles (TT). Heterozygous means contrasting alleles (Tt). Phenotype is the observable character (e.g., tall). $Filial_1$ is the first generation progeny.
Final Answer : (1)

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156) **Explanation:** According to the Punnett square for a dihybrid cross, the genotype Rryy occurs twice out of the 16 possible combinations.
Final Answer : (2)

157) **Explanation:** *Drosophila* is suitable because there is a clear differentiation of the sexes; male and female flies are easily distinguishable.
Final Answer : (3)

158) **Explanation:** In standard human pedigree analysis, a diamond symbol is used when the sex of the individual is unspecified.
Final Answer : (2)

159) **Explanation:** An individual affected by Down's syndrome has a short stature with a small round head, furrowed tongue, and partially open mouth.
Final Answer : (3)

160) **Explanation:** Honey bees have a haplodiploid sex-determination system where females are diploid (32 chromosomes) and males are haploid (16 chromosomes).
Final Answer : (3)

161) **Explanation:** Turner's syndrome is caused by the absence of one of the X chromosomes, leading to a karyotype of 45 with XO.
Final Answer : (3)

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162) **Explanation:** Henking (1891) could trace a specific nuclear structure (the X body) all through spermatogenesis in a few insects.
Final Answer : (4)

163) **Explanation:** A single gene exhibiting multiple phenotypic expressions is called a pleiotropic gene, with Phenylketonuria being a classic example.
Final Answer : (2)

164) **Explanation:** If starch grain size is considered as the phenotype, the intermediate size in heterozygotes (Bb) illustrates incomplete dominance.
Final Answer : (3)

165) **Explanation:** A son inherits his X chromosome exclusively from his mother. Since the mother is homozygous normal (XX), all sons will receive a normal X chromosome and will not be colour blind.
Final Answer : (1)

166) **Explanation:** Alfred Sturtevant, a student of Morgan, used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome.

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Final Answer : (3)

167) **Explanation:** For a child to have blood group O (genotype ii), both parents must contribute an 'i' allele. Therefore, both the father (blood group A) and the mother (blood group B) must be heterozygous (I^A i and I^B i respectively).
Final Answer : (2)

168) **Explanation:** The Law of Segregation is based on the fact that the alleles do not show any blending and that both characters are recovered as such in the F_2 generation, though one of these is not seen in the F_1 stage.
Final Answer : (2)

169) **Explanation:** While Sutton and Boveri proposed the chromosomal theory of inheritance, the experimental verification of this theory was carried out by Thomas Hunt Morgan and his colleagues using *Drosophila melanogaster*.
Final Answer : (2)

170) **Explanation:** The number of gametes produced by a heterozygous organism is calculated by the formula 2^n, where n is the number of heterozygous loci. Here, n = 4. So, $2^4 = 16$.
Final Answer : (3)

171) **Explanation:** Phenylketonuria is an inborn error of metabolism where the affected individual lacks an enzyme that converts the amino acid phenylalanine into tyrosine.
Final Answer : (1)

172) **Explanation:** Klinefelter's syndrome occurs due to the presence of an additional copy of the X-chromosome resulting in a karyotype of 47, XXY. Such an individual has overall masculine development, but feminine development (gynaecomastia) is also expressed, and they are sterile males.
Final Answer : (2)

173) **Explanation:** Haemophilia is a sex-linked (X-linked) recessive disease, which shows transmission from an unaffected carrier female to some of the male progeny.
Final Answer : (3)

174) **Explanation:** A cross between an F_1 progeny and a homozygous recessive parent is called a test cross, and it is used to determine the genotype of an organism showing a dominant phenotype.
Final Answer : (2)

175) **Explanation:** Morgan found that the genes for white and miniature wing were loosely linked and showed 37.2 per cent recombination.
Final Answer : (2)

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176) **Explanation:** The woman inherited the colour-blind X chromosome (X^c) from her father, making her a carrier (X^c X). The normal man has an XY genotype. Sons inherit their X chromosome from their mother. There is a 50% chance the mother passes the X^c chromosome and a 50% chance she passes the normal X chromosome. Thus, 50% of their sons will be colour blind.
Final Answer : (3)

177) **Explanation:** Mutation that arises due to a change in a single base pair of DNA is known as a point mutation. A classic example of such a mutation is sickle cell anaemia.
Final Answer : (2)

178) **Explanation:** Out of the three possible genotypes, only homozygous individuals for Hb^S (Hb^S Hb^S) show the diseased phenotype. Heterozygous individuals (Hb^A Hb^S) appear apparently unaffected but they are carriers of the disease.
Final Answer : (3)

179) **Explanation:** In birds, females are heterogametic and produce two different types of eggs (Z and W), while males are homogametic producing only one type of sperm (Z). Therefore, the sex of the offspring is determined by the type of egg (Z or W) that gets fertilized by the Z-bearing sperm.
Final Answer : (3)

180) **Explanation:** Traits that are generally controlled by three or more genes are called polygenic traits. Human skin colour is a classic example of polygenic inheritance, influenced by multiple genes.
Final Answer : (4)

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